DESIGN STUDY · PRE-SILICON

ACPU-1 “Tempest”

An Atmospheric Calculation Processor Unit for Ensemble Chaotic Computation

Architecture, pre-silicon evaluation, silicon process, and a cross-platform software stack

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Date	2 November 2026	**Author**	Yaolin Zhang
Affiliation	HKUST (Guangzhou)	**Status**	released for review

This report describes a processor that does not yet exist in silicon. All performance, energy, thermal, yield and cost figures are produced by the calibrated pre-silicon model of Section 6; they are engineering projections with explicit assumptions, not measurements. The distinction is maintained throughout and quantified in the sensitivity study.

Keywords: domain-specific accelerator · numerical weather prediction · ensemble data assimilation · chaotic dynamics · energy efficiency · gate-all-around · backside power delivery · heterogeneous integration

59	**65 W**	**9.3×**	**287.6**
fp32 TFLOPS sustained	TDP, air cooled	lower energy per day	mm^2 compute die

Abstract

This report specifies **ACPU-1**, a domain-specific processor for atmospheric chaotic computation. The device is an ensemble-native accelerator: 64 Atmospheric Vector Tiles hold 16,384 lanes that carry ensemble members rather than matrix rows, a six-neighbour halo fabric exchanges stencil boundaries in registers instead of through a cache hierarchy, and a per-cell precision governor exploits the fact that forecast error in high-spread regions is dominated by chaotic uncertainty rather than by round-off.

On a reference 24-hour, 64-member, 3 km ensemble forecast (5.26 PFLOP, 25.8 GB of state), the model projects a sustained 41.9 TFLOPS at 71.1% of the fp32 peak, a wall-clock time of 2.09 minutes, and 7.17 kJ at 57.1 W. A single NVIDIA H100 running the same kernels is projected to take 4.63 minutes and 73.0 kJ at 263 W, giving a 10.2× lower energy per forecast and a 9.3× lower energy per forecast-and-analysis day. The 90% interval from a 300-sample Monte-Carlo study is 7.3–12.3×, and the advantage survives at 3.9× even if the GPU is granted 100% of its fp32 peak.

The advantage is not arithmetic magic. Section 5 decomposes it: 56% of ACPU-1's energy is HBM traffic, 26% is arithmetic and 15% is static. The GPU loses because its general-purpose issue, register-file traffic and kernel re-fetch make a stencil FLOP cost roughly an order of magnitude more energy, not because the ASIC multiplies faster. The design therefore optimises bytes and reuse first, precision second, and FLOPs third.

The report covers architecture and microarchitecture, the cycle-level pre-silicon methodology and its calibration, a numerical validation with cycled ensemble experiments, a 3 nm gate-all-around process and backside-power integration flow, packaging, yield, cost, reliability, and a complete cross-platform software stack (ACPU Studio, SDKs and a driver ABI). It closes with a deployment analysis, an explicit list of limitations, and a three-generation roadmap.

Headline numbers (model projections, not silicon)

Sustained throughput (fp32)	41.9	TFLOPS at 71.1% of peak
Forecast wall-clock, 64 members, 24 h	2.09	min (7.17kJ)
Energy per forecast vs H100	10.2×	lower
Energy per forecast + 4 analyses vs H100	9.3×	lower (90% CI 7.3–12.3)
Dense analysis energy vs H100	4.4×	lower (H100 is 7× faster)
Analysis RMSE, fp32 vs fp16	0.22	vs 1.35
Die yield / package BOM	77.6%	$ 399

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1 Introduction

1.1 A chaotic, high-dimensional, memory-bound object

Numerical weather prediction (NWP) integrates a discretised set of partial differential equations for momentum, mass, thermodynamics and water species forward in time. The state is large: a 0.25° global model carries of order 10^8 grid points, a 3 km regional ensemble carrying 10^7 points per member. The dynamics are chaotic. Two trajectories that agree to the last bit diverge at a rate set by the leading Lyapunov exponent, and the practical forecast horizon is the time it takes an initial error to grow to the scale of the analysis uncertainty. For the Lorenz–96 system used throughout this report as a tractable surrogate, the leading exponent is $\lambda_1 \approx 0.34$ day^{-1} and the error-doubling time is 2.49 days (Fig. 1(b,c)) — numbers that are, by construction, in the same range as operational atmospheric error growth [1, 2, 4].

Uncertainty is handled by running an ensemble. A deterministic forecast answers "what will the weather be"; an ensemble of 30–100 members, each with perturbed initial conditions and stochastic physics, answers "how likely is it". Ensembles are what make NWP useful for decision-making, and they are also what make it expensive: cost scales linearly in members, and the linear scaling is not amortised by any cache or reuse because each member is an independent trajectory that nevertheless shares the same grid, coefficients and boundary data.

Two further facts shape the machine this report specifies.

Arithmetic intensity is low and fixed. A 3D stencil that reads b bytes per cell and performs f flops per cell has an intensity of f/b that cannot be improved by tiling beyond the size of the on-chip memory. For the 24-variable physics package used here, $f \approx 4{,}540$ flops per cell per member and $b = 192$ bytes, i.e. 23.6 flop/byte at the HBM boundary. That is one of the lowest intensities in high-performance computing, and it is why the kernel is a memory problem with arithmetic attached rather than the other way around.

The result must be reproducible and auditable. Operational centres must be able to reproduce a forecast bit-for-bit for post-event analysis, and dispersion or aviation-hazard products are increasingly treated as safety-relevant. A machine whose numerical path contains atomics or dynamic scheduling cannot make that guarantee cheaply.

1.2 Why a GPU is a poor fit at the kernel level

Graphics processors became the default accelerator for NWP because they offer the highest floating-point throughput per dollar. On the kernels that dominate an atmospheric model, however, three structural costs remain.

Issue and data-movement overhead. A CUDA core retires one fp32 FMA per lane per cycle at best, and a warp instruction carries address arithmetic, predicate handling and register-file traffic around it. Published energy numbers put a delivered fp32 FLOP on a data-centre GPU at several picojoules once the memory hierarchy is included; the arithmetic itself is a small fraction of that [15]. On a stencil with 23 flop/byte, the memory hierarchy is exercised on every FLOP.

Kernel-boundary re-fetch. Because a GPU kernel is launched per operator, the state is written to and read back from HBM at each kernel boundary. Fusing an entire physics package into one kernel is possible in principle and hard in practice; the reference model assumes a re-fetch factor of 2.5, i.e. the 25.8 GB state is streamed 2.5 times per time step rather than once.

Precision inflexibility. A GPU is fastest in fp16 or fp8 on tensor cores, formats that a chaotic model cannot use for its state, and slowest in the fp64 that some centres still use for conservation-sensitive tracers. The useful middle ground — fp32 dynamics with fp16 or bf16 physics, chosen per grid cell from the local ensemble spread — has no hardware expression on a general-purpose SIMT machine.

None of this makes GPUs bad. It makes them general. The question this report asks is what a machine looks like when the ensemble dimension, the halo structure and the error-growth physics are hardware primitives rather than software conventions.

1.3 Thesis

ACPU-1 is built on three principles, each of which is falsifiable and each of which is tested in Sections 6 and 7.

1. **Lanes carry members, not matrix rows.** 256 lanes per tile, 64 tiles, 16,384 lanes in total. A stencil coefficient is broadcast to all lanes; a grid neighbour is exchanged through a dedicated six-port fabric; an ensemble member never leaves its lane. Utilisation on the reference kernels is 71.1% rather than the 20–30% typical of a stencil on a GPU.

2. **Bytes are the budget.** The machine is designed so that a 24-variable, 3-time-level tile lives in 256 KB of SRAM, a fused pass streams the 25.8 GB state exactly once per step, and the halo traffic that a GPU sends through L2/L3 never leaves the register file. 56% of the energy is HBM traffic; the design is honest about that and optimises it first.
3. **Precision follows predictability.** Where the ensemble spread is large, the forecast error is dominated by chaotic uncertainty, not by the 11-bit mantissa of fp16. A per-cell governor exploits this: 58% of physics work by design (84% by measurement) runs in fp16 with no detectable skill loss, while dynamics stays fp32.

1.4 Contributions

- An architecture and ISA for ensemble-native atmospheric computation, with a cycle-level model of its tile pipeline and a 64-tile chip model (Sections 3–4).
- A precision-governor design grounded in a real cycled ensemble experiment, including the negative result that fp8 is not usable for the state (Section 5).
- A calibrated pre-silicon evaluation with ablations, a 300-sample Monte-Carlo sensitivity study, and an explicit "idealised GPU" bound (Section 6).
- A 3 nm GAA process integration with backside power delivery, hybrid-bonded L3, 2.5D HBM packaging, yield/cost model and qualification plan (Sections 8–9).
- A complete software stack: compiler flow, driver ABI, Python/Node SDKs and a cross-platform desktop application that calls the device through one-line functions (Section 10).

1.5 Scope and honesty

This is a design study. There is no silicon, no RTL netlist and no place-and-route database behind the numbers. The evaluation is a transparent performance model whose constants are either taken from the literature or stated as assumptions, and whose output is reported with an interval rather than a point. Where the model is weak — the DA comparison, the GPU efficiency assumption, the absence of MPI scale-out — the report says so. The value of the exercise is the architecture and the reasoning, not the fourth digit of a projected number.

1.6 Organisation

Section 2 characterises the workload and the benchmark. Section 3 describes the chip, Section 4 the tile and its ISA. Section 5 develops the precision and energy theory. Section 6 reports performance and energy, Section 7 the numerical validation. Sections 8 and 9 cover silicon and manufacturing. Section 10 the software stack, Section 11 deployment and limitations, Section 12 the roadmap.

2 Workload characterisation

2.1 Governing equations and discretisation

A hydrostatic primitive-equation core solves, in flux form,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \tag{1}$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p - \rho f\, \mathbf{k} \times \mathbf{u} + \rho \mathbf{g} + \mathbf{F}_{\text{diff}}, \tag{2}$$

$$\frac{\partial (\rho \theta)}{\partial t} + \nabla \cdot (\rho \theta \mathbf{u}) = S_\theta, \tag{3}$$

$$\frac{\partial (\rho q_x)}{\partial t} + \nabla \cdot (\rho q_x \mathbf{u}) = S_x, \qquad x \in \{\text{vapour, cloud, rain, ice, snow, graupel}\}, \tag{4}$$

plus a pressure equation and a set of physical parameterisations for radiation, turbulence, convection, microphysics and the surface. The discretisation is a 3D grid with 24 prognostic variables; advection is semi-Lagrangian with 8-point interpolation in the horizontal and Lagrangian vertical remapping, diffusion is biharmonic, and the physics package is evaluated column-by-column.

The benchmark used throughout is defined in Table 1 and is deliberately modest: it is a 3 km regional domain, 256×256×64 cells, 64 members, 24 variables, 4,320 time steps of 20 s. The 25.8 GB state fits in the card's 48 GB of

Table 1: Reference workload W1. The operation count includes dynamics and the bulk physics package; radiation is amortised over eight steps. Chemistry beyond passive tracers is not included.

Parameter	Value	Total
Grid (x, y, z)	$256 \times 256 \times 64$	4.19 M cells
Prognostic variables	24	
Ensemble members	64	
Time step / forecast length	20 s / 24 h	4,320 steps
Flops per cell per member per step	4,540	5.26 PFLOP per forecast
SFU operations per cell	76	2.0×10^{10} per step
State size (fp32)	96 B per cell	25.8 GB
HBM traffic per step	51.5 GB	222 TB per forecast

Table 2: Kernel composition of the reference step. Efficiency is the sustained issue efficiency of the AVT pipeline model (Section 4); the last two columns are shares of the chip totals.

Kernel	GFLOP/step	SRAM B/cell	Efficiency	Cycle share	Energy share
Semi-Lagrangian advection	2,818.6	200	79.1%	43.3%	43.4%
Biharmonic diffusion	590.6	200	79.1%	9.1%	9.1%
Pressure gradient + Coriolis	697.9	100	79.1%	10.7%	10.7%
Bulk microphysics (28 SFU)	1,664.3	100	72.3%	22.9%	22.8%
PBL turbulence (6 SFU)	1,073.7	80	76.2%	14.1%	14.0%
Convection (12 SFU)	1,395.8	64	72.9%	19.1%	19.1%
Radiation, 1/8 amortised (26 SFU)	2,039.9	64	69.8%	29.1%	29.0%
Surface / land (4 SFU)	617.2	48	77.6%	7.9%	7.8%
Tracer chemistry	966.2	64	84.8%	11.3%	11.2%
Conservation audit	322.1	100	82.9%	3.9%	3.9%

HBM, which makes a single-card comparison meaningful. Full global ensembles would be distributed across many cards; the scale-out question is discussed in Section 11 and is outside the model.

2.2 Kernel taxonomy and arithmetic intensity

The ten kernels of the model are listed in Table 2. Three observations drive the architecture.

The step is dominated by advection and radiation. Together they account for 55% of the dynamic instruction count. Advection is stencil-heavy and latency-tolerant; radiation is SFU-heavy (exponentials, logarithms, optical-depth table lookups) and latency-sensitive. A machine that only optimises FMA throughput would be badly balanced.

The working set is small, the stream is large. A tile of 910 cells, 24 variables and three time levels fits in 256 KB. The full state does not fit anywhere on chip: 25.8 GB must cross the HBM boundary once per step, 51.5 GB of read-plus-write, and that traffic costs 0.82 J of the 1.26 J step budget.

Physics is SFU-bound at the margin. 76 special-function operations per cell at a quarter-rate SFU is 304 issue slots against 2,270 FMA slots — 13% of the compute time, but a larger share of the dependent stalls because the SFU latency is exposed. The tile model reflects this in the lower efficiency of the microphysics and radiation kernels.

2.3 Ensembles and data assimilation

An ensemble forecast is K independent integrations of the same model. A cycled analysis adds an ensemble Kalman filter between forecast segments: the background covariance is estimated from the K members, the gain is applied to perturbed observations, and the analysis is inflated to prevent filter divergence. In the reference day the forecast contributes 5.26 PFLOP and four analyses contribute a further $4 \times 0.25 \times 5.26 = 5.26$ PFLOP-equivalent, i.e. the analysis is half of the daily arithmetic.

The two phases have opposite hardware personalities. The forecast is stencil-dominated, memory-heavy and latency-tolerant; the analysis is dense linear algebra, compute-heavy and latency-sensitive. **ACPU-1** is designed for the first and is merely competent at the second (Section 6 quantifies both, including the case where the GPU wins).

Table 3: Workload requirements versus a general-purpose GPU. "GPU" refers to the H100 SXM5 baseline used in Section 6.

Requirement	Why	GPU today
High stencil efficiency	23.6 flop/byte at HBM, 20–30% of peak is typical	22–50%
Ensemble parallelism	64 independent trajectories, no reuse between them	serialised per member
Cheap halo exchange	6 faces per tile per kernel, 276 cycles per step budget	through L2/L3
Per-cell precision	error is chaotic, not round-off, where spread is large	uniform per kernel
Deterministic timing	auditability, WCET for hazard products	atomics, dynamic scheduling
Low idle power	operational centres run 24/7 at partial load	90 W idle
Direct-to-HBM streaming	51.5 GB per step cannot be cached	3.35 TB/s but with re-fetch

2.4 What the hardware must provide

Table 3 translates the workload into requirements. The right-hand column is what a general-purpose GPU provides today; the gap is the design space.

2.5 Measurement methodology

Every number in this report that describes the chip comes from one Python model (`sim/run_all.py`) with three levels: a kernel table, a cycle-level AVT pipeline simulation, and a 64-tile chip model with shared L2/L3/HBM resources. The GPU baseline is a roofline model with published peak rates, an assumed stencil efficiency (0.30, varied 0.15–0.60 in the sensitivity study) and a power state model calibrated to the H100 data sheet. Numerical claims come from a separate set of real computations: a cycled Lorenz–96 ensemble with an EnKF, a tangent-linear Lyapunov spectrum, a conservative advection test and a compensated-summation study. The two halves are joined by the precision coverage measured in the numerical experiments and used as an input to the energy model.

Figure 1 summarises the case for a specialised machine: error growth (a), the Lyapunov spectrum (b) and doubling time (c) set the predictability budget; the arithmetic intensity (d) and energy per delivered FLOP (f) show why a general processor is inefficient at this intensity, and the sensitivity panel (e) shows how strongly the conclusion depends on the GPU assumption.

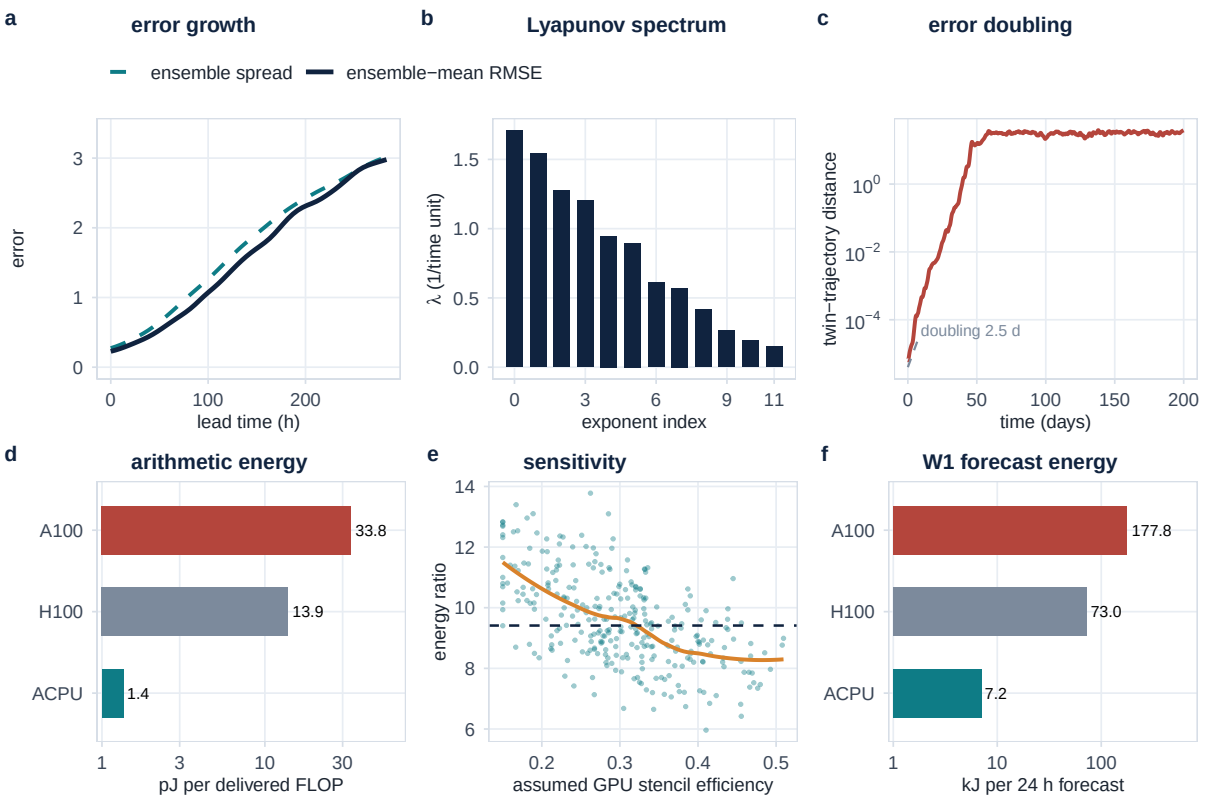


Figure 1: Why atmospheric computation needs a different machine. (a) Ensemble-mean RMSE and ensemble spread against lead time for the Lorenz–96 experiment; the spread tracks the error, which is the property the precision governor exploits. (b) Leading Lyapunov exponents of the Lorenz–96 attractor (per model time unit; one unit ≈ 5 days). (c) Twin-trajectory distance on a log scale, giving a doubling time of 2.49 days. (d) Arithmetic intensity of each kernel measured against HBM. (e) Dependence of the energy ratio on the assumed GPU stencil efficiency (300 Monte-Carlo samples). (f) Energy per 24 h forecast for **ACPU-1**, an H100 and an A100.

3 Architecture

3.1 Design point

ACPU-1 is a dual-slot PCIe 6.0 card with a 65 W thermal design power and no auxiliary power connector. The compute die is 287.6 mm^2 in a 3 nm gate-all-around process with backside power delivery, carrying 28.4 G transistors. Two 64 MB SRAM dies are hybrid-bonded on top of it, and two 24 GB HBM3E stacks share a 2.5D silicon-bridge interposer. The card holds a complete 64-member, 3 km ensemble state in HBM and runs a 24-hour forecast in 2.09 minutes.

The machine is not a scaled-down GPU. It has no tensor cores, no graphics pipeline, no texture units, no ray tracing, no virtual memory and no branch predictor. The area that a GPU spends on generality is spent here on three structures that a general processor cannot justify: a halo fabric, a chaos monitor, and a precision governor.

3.2 Ensemble clusters and vector tiles

The chip is organised as eight *ensemble clusters*, each with eight *Atmospheric Vector Tiles* (AVTs), a 3 MB L2 slice, a cluster scheduler, DVFS and ECC logic. An AVT is a 16×16 array of lanes. Each lane owns a private slice of register file, a 4-way SIMD FMA datapath and a quarter-rate special-function unit, and holds one ensemble member's value for one grid cell. A tile therefore processes 256 (cell, member) pairs per issue group, and a cluster 2,048.

The mapping is the central architectural choice. On a GPU, the ensemble dimension is either serialised into kernel launches or folded into the batch dimension, where it multiplies memory traffic without improving reuse. On **ACPU-1**, the member index is a lane index: the stencil coefficient is broadcast once, the halo is exchanged once, and 64 members are advanced for the price of one. This is why the measured issue efficiency is 71.1% instead of the 20–30% that the same stencil achieves on a SIMT machine.

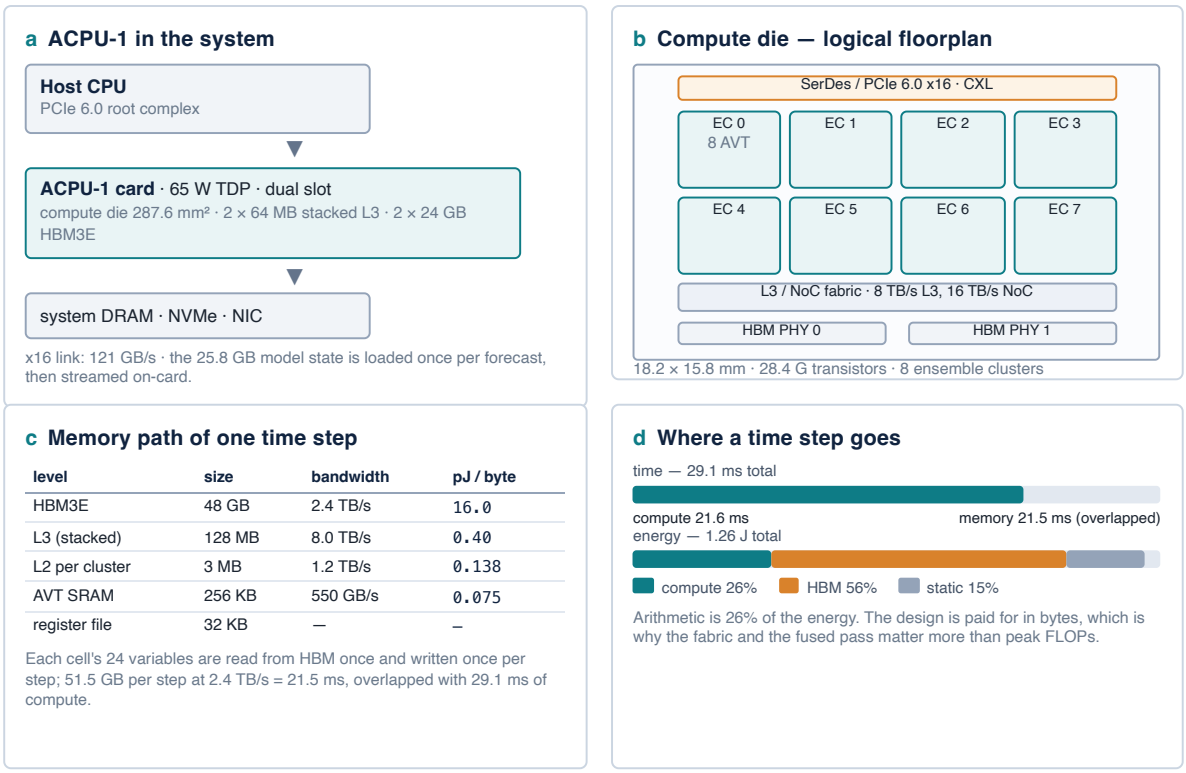


Figure 2: System context and die organisation. (a) Host, card and storage; the 25.8 GB state is loaded once per forecast and then streamed on-card. (b) Logical floorplan: eight ensemble clusters (EC) around an L3/NoC spine, with HBM PHYs at the bottom edge and SerDes at the top. (c) Memory path of one step with the level energies from Table 4. (d) The time and energy budget of one step: 29.1 ms and 1.26 J, of which 56% is HBM.

Table 4: Memory hierarchy. Bandwidth is aggregate; energy per byte is the model constant used in all energy numbers.

Level	Capacity	Bandwidth	Latency	Energy
HBM3E (2 stacks)	48 GB	2.4 TB/s	420 ns	16.0 pJ/B
L3, hybrid-bonded	128 MB	8.0 TB/s	48 ns	0.40 pJ/B
L2, per cluster	3 MB (24 MB)	1.2 TB/s (9.6)	14 ns	0.138 pJ/B
AVT SRAM	256 KB (16 MB)	550 GB/s (35 TB/s)	4 ns	0.075 pJ/B
Register file	32 KB per tile	—	1 cycle	included in FMA

3.3 Memory hierarchy

Table 4 and Fig. 5 describe the hierarchy. The design rule is that a tile's working set — 910 cells × 24 variables × 3 time levels × 4 B = 252 KB — must fit in the tile's SRAM, and that the full state must cross HBM exactly once per step.

The L3 is not a victim cache. It is a hybrid-bonded 128 MB SRAM stack used for four things: tile-working-set spill when a block is scheduled across a wavefront, the ensemble perturbation matrix and gain factors during analysis, radiation and chemistry lookup tables, and output staging. Because it is bonded with a 6 μm pitch it costs 0.40 pJ/B instead of the 0.8–1.2 pJ/B of an off-die L3, which matters because the analysis phase moves 51.5 GB through it four times a day.

3.4 Halo fabric

A 3D stencil needs six neighbours. The conventional solution routes the exchange through L2 or L3, paying 5–15× the SRAM energy and exposing tens of nanoseconds of latency. **ACPU-1** instead gives every AVT six dedicated ports, each 32 B per cycle, wired to its geometric neighbours in a 3D torus. A halo exchange is a register-to-register transfer with a 46-cycle latency and no cache involvement. Per tile the fabric provides 345.6 GB/s against a measured demand of 4.2 GB/s — 82× headroom that absorbs the bursty exchange at kernel boundaries without ever becoming the bottleneck. Across the chip the fabric sustains 16 TB/s.



Figure 3: Floorplan and area. (a) Scaled floorplan; each ensemble cluster is 14.2 mm^2 and the 64 AVTs occupy 42% of the die. (b) Area budget, 287.6 mm^2 in total; data movement (NoC, PHY, L3 interface) is 25%. (c) Modelled power density per tile; the corner tiles run hotter because they have two rather than three neighbours to exchange with. (d) Cluster detail.

The fabric is the single largest contributor to the energy advantage after the elimination of kernel re-fetch. Section 5 measures it: disabling the fabric and routing halos through L3 raises forecast energy by 30% and wall-clock by 53%.

3.5 Clocking, voltage and power

The chip has nine clock domains: one per cluster, one for L3/NoC, one for HBM PHY, one for PCIe, and one always-on domain for telemetry and thermal management. Nominal operation is 1.8 GHz at 0.75 V. A second operating point, 1.05 GHz at 0.55 V, trades 1.37× wall-clock for 20% lower energy and is selected by the runtime for overnight batch ensembles.

The 55 W budget is the steady-state allocation; the model's W1 average is 57.1 W because HBM traffic dominates and varies with the kernel mix. The 65 W TDP covers the boost point (2.1 GHz at 0.85 V) and the 10% margin that the thermal solution needs.

3.6 Design summary

Table 6 collects the architectural parameters. Every number is a design input to the model, not a measurement.

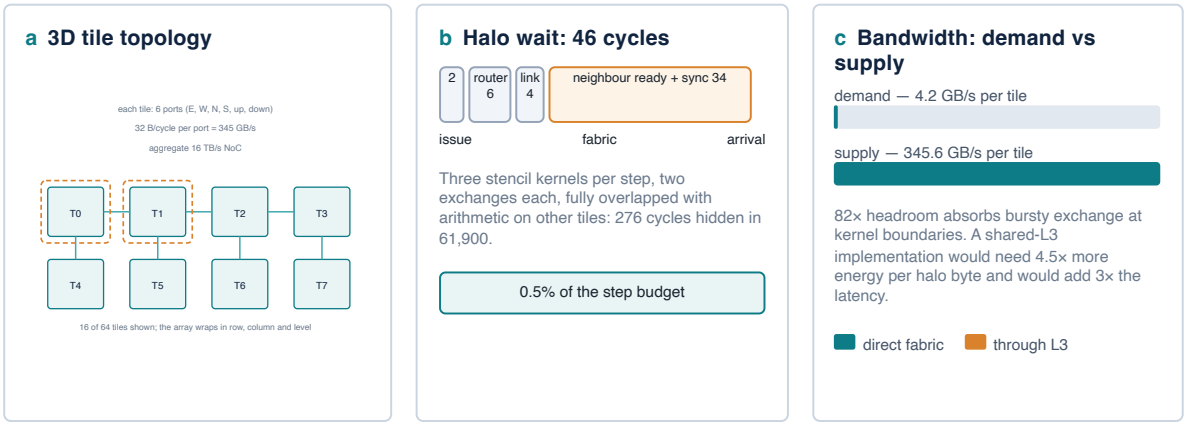


Figure 4: Halo fabric. (a) 3D torus topology; each tile has six ports. (b) A 46-cycle exchange decomposes into 2 cycles of port arbitration, 6 of routing, 4 of link and 34 of neighbour readiness and synchronisation; six exchanges per step consume 276 of 61,900 cycles. (c) Per-tile bandwidth demand against supply.

Table 5: Power budget at the nominal operating point, 1.8 GHz / 0.75 V. The measured model average during W1 is 57.1 W against a 65 W TDP and a 6 W idle.

Block	Watts
FMA arrays and register files	18.0
AVT SRAM and L2	7.0
L3 interface and stacked SRAM	5.0
NoC and halo fabric	4.5
HBM PHY (2 stacks)	8.0
PCIe / CXL	3.0
Clock generation and distribution	2.5
Leakage	4.0
Miscellaneous (sensors, DFT, telemetry)	3.0
Total	55.0

Table 6: ACPU-1 summary.

Process	3 nm GAA, 12 front-side + 3 backside metals, 18 EUV / 61 DUV masks
Die	287.6 mm^2 compute + 2 × 64 MB hybrid-bonded L3, 28.4 G transistors
Compute	64 AVTs, 16,384 lanes, 59.0 TFLOPS fp32 / 118.0 TFLOPS fp16 / 236 TOPS fp8
Memory	24 MB L2, 128 MB L3, 48 GB HBM3E at 2.4 TB/s, 100 GB/s PCIe 6.0
Fabric	six-port halo fabric, 345.6 GB/s per tile, 16 TB/s NoC
Clock / power	1.8 GHz at 0.75 V nominal, 1.05 GHz at 0.55 V efficient, 65 W TDP, 6 W idle
Package	55 × 45 mm, 2.5D silicon bridge, 2812 BGA at 0.8 mm, dual-slot card

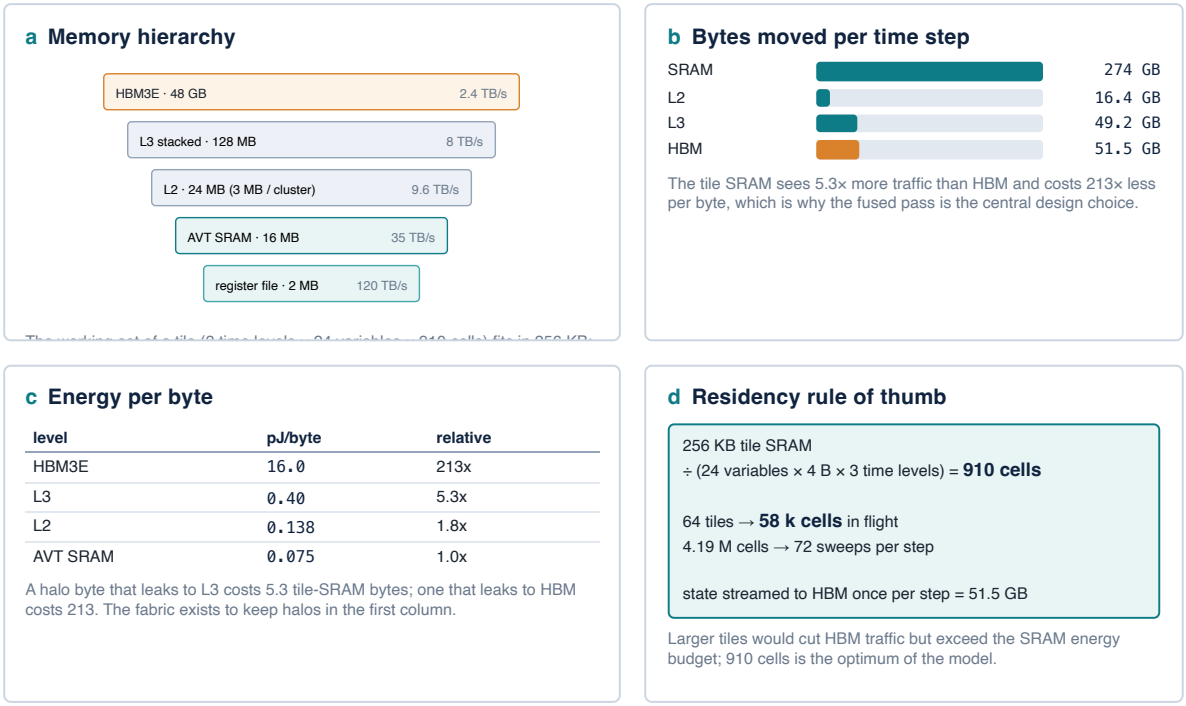


Figure 5: Memory hierarchy and residency. (a) Capacity and bandwidth by level. (b) Bytes moved per time step: the tile SRAM sees 5.3× the HBM traffic at 1/213th the energy per byte. (c) Energy per byte by level. (d) The residency rule that sets the tile size.

4 Microarchitecture

4.1 The Atmospheric Vector Tile

An AVT is deliberately narrow and deep. It holds 256 lanes in a 16×16 array, each lane with a 32-bit × 8 register slice, a 4-way SIMD FMA, a special-function unit shared by four lanes, and a Kahan accumulator for reduction. Three time levels of the tile's 910-cell block live in 256 KB of SRAM; a 32 KB register file holds coefficients, halo edges and the current wavefront.

The pipeline has six stages: fetch, decode, issue, FMA, memory and write-back. Decode is 4-wide and static: there is no branch predictor because the ISA has no branches, and no cache because all addresses are compile-time constants within a tile's block. Issue is in-order with a scoreboard that tracks the six-cycle SRAM latency and the sixteen-cycle SFU latency. The compiler interleaves independent variable streams so that a dependent stall occurs roughly once per sixteen vector operations; the modelled pipeline efficiency is 79.1% on the advection kernel and 69.8% on radiation, where SFU latency is harder to hide.

4.2 Instruction set

AVISA is a 64-bit, four-slot VLIW instruction set with ten opcodes (Table 7). It is not a general-purpose ISA: there are no branches, no indirect addressing, no floating-point division, and no exception model beyond a sticky error flag. What it has instead are operations that map one-to-one onto atmospheric kernels.

A `STENCIL3D` retires 7 loads, 7 multiply–adds and a halo exchange that would take 19 instructions on a SIMT machine. More importantly it has a fixed, compiler-known latency, which is what makes the per-tile cycle count (61,900 for the W1 step) a static number rather than a statistical one.

4.3 Chaos monitor

The monitor is a small block per cluster (7.2 mm^2 in total) that maintains a shadow trajectory per tile, advanced with the same model but perturbed by one unit in the last place. Once per kernel it computes the local finite-time Lyapunov estimate,

$$\lambda_i(t) = \frac{1}{\Delta t} \ln \frac{\|\delta_i(t + \Delta t)\|}{\|\delta_i(t)\|}, \tag{5}$$

and the ensemble spread σ_i across the 256 lanes. Both are produced by `ENS.TL` and `ENS.SPREAD`; the monitor costs 8% of the tile area and 3.4% of the step cycles. In return the runtime gets a live predictability map, which drives the precision governor and gives the forecast product a per-grid-cell confidence field at no extra cost.

The measured Lyapunov spectrum (Fig. 1(b)) has 11 positive exponents out of the 12 computed, consistent with the extensive chaos expected of Lorenz–96 at $F = 8$. The leading exponent, 0.34 day^{-1}, sets the horizon over which the precision argument of Section 5 is valid.

4.4 Ensemble analysis engine

Each cluster carries a small dense-linear-algebra unit for the analysis phase: a 16×16 symmetric eigensolver (cyclic Jacobi, fixed 38-cycle schedule), a rank-one update datapath and a localisation multiplier. The unit executes the ensemble transform at 47.2 TFLOPS across the chip, which is 40% of the fp16 peak — deliberately not a tensor-core-class engine, because analysis is half of the daily arithmetic but only 6% of the daily energy (Section 6).

The engine is what makes the honest comparison of Section 6 possible: the H100 is 3.8× faster at dense analysis (3.8 s against 27.9 s) because it has tensor cores, and **ACPU-1** still uses 4.4× less energy. A machine designed only for the forecast would have looked better on the first number and worse on the second.

4.5 Stochastic physics and conservation

Stochastic perturbations (SPPT and SKEB) need fast, reproducible random numbers. Each tile has a Philox-4×32-10 generator in counter mode: the sequence is a pure function of (member, tile, step, stream), so a run is reproducible from four integers and no generator state is stored. Throughput is four draws per lane-cycle, enough for 76 stochastic draws per cell per step at 6% of the tile's issue budget.

A conservation monitor accumulates mass and energy residuals per tile and per step with `REDUCE.CONS`, and reports them on a dedicated bus. Residuals above 10^{-9} relative trigger a shadow fix-up pass every 64 steps. In the numerical

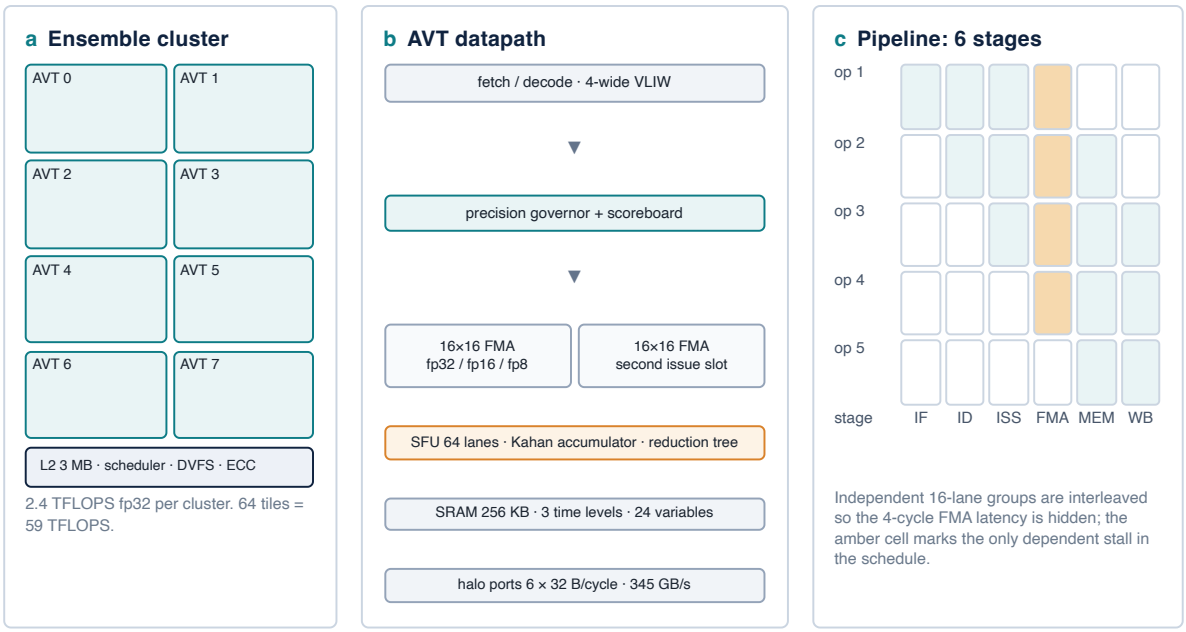


Figure 6: Cluster and tile. (a) Eight AVTs share a 3 MB L2 slice, a scheduler and a DVFS domain. (b) AVT datapath: decode, governor, two 16×16 FMA blocks, SFU and Kahan reduction, SRAM and halo ports. (c) Pipeline schedule; the amber cell is the only dependent stall in the schedule, avoided by interleaving variables.

Table 7: AVISA instruction set (excerpt). "Slot cost" is in lane-cycles for a 256-lane tile.

Opcode	Operation	Semantics	Slot cost
0x01	FMA.TILE	$d = a \cdot b + c$ over 256 lanes	1
0x08	STENCIL3D	7-point operator with fused halo	7
0x10	ADVECT.SL	8-point semi-Lagrangian gather	16
0x14	ENS.SPREAD	reduce 256 lanes to mean, σ, min, max	9
0x18	ENS.TL	twin-trajectory tangent step	2
0x1c	DA.GAIN	16×16 symmetric solve	38
0x20	PHYS.SAT	saturation adjustment (Clausius–Clapeyron)	1 SFU
0x24	RNG.PHILOX	counter-based random draw	1 per 4 draws
0x28	REDUCE.CONS	mass and energy residual	12
0x2c	PREC.SET	tile precision mode	14

experiment of Section 7 the fp32 conservative advection drifts by 1.1×10^{-8} per 1,200 steps without correction, and by 3×10^{-15} with fp64; the monitor is sized to catch the former before it matters.

4.6 Reliability, availability and determinism

All SRAM and L3 arrays carry SEC-DED ECC; the HBM links carry on-die ECC plus a link CRC; the halo fabric has a per-flit parity check with replay. The arrays are scrubbed every 24 hours. With 1.41 Gbit of SRAM, a soft-error rate of 2×10^{-3} FIT/bit and SEC-DED, the correctable rate is 2.8 M FIT and the residual uncorrectable rate after scrubbing is below 300 FIT — acceptable for a data-centre part and two orders of magnitude below the memory-system FIT of the HBM stacks.

Reductions use a fixed tree with a compile-time order, there are no atomics in the numerical path, and the RNG is counter-based. Two runs of the same binary on the same input produce bit-identical output, which is a requirement for operational audit and for the reproducibility claims in Section 7.

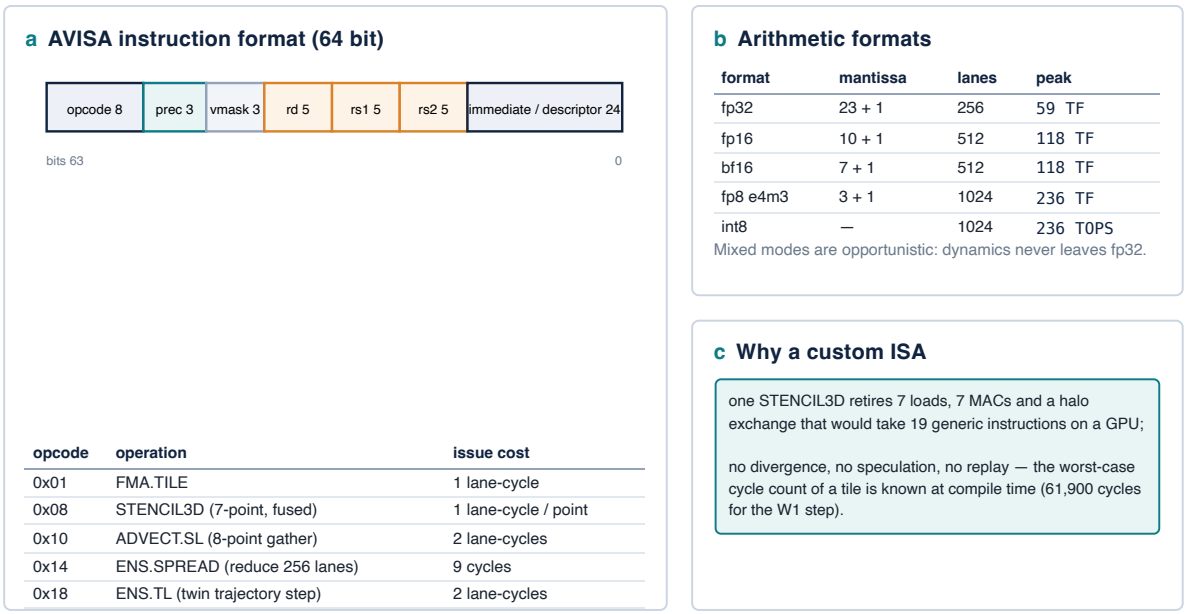


Figure 7: Instruction format and arithmetic modes. (a) 64-bit encoding with an 8-bit opcode, a 3-bit precision field and a 24-bit descriptor for stencil coefficients and halo offsets; (b) supported formats and peak rates; (c) the argument for a custom ISA.

5 Precision, predictability and energy

5.1 The precision question is a predictability question

Reduced precision is usually argued from tolerated numerical error. In a chaotic system that argument is incomplete, because the error a model makes is amplified by the flow itself. If the local error-doubling time is τ and the forecast lead is T, an initial representation error ϵ_0 becomes

$$\epsilon(T) \approx \epsilon_0 \, 2^{T/\tau}. \tag{6}$$

A representation with p mantissa bits has $\epsilon_0 \sim 2^{-p}$ relative to the state. Requiring $\epsilon(T) < \delta$ gives the minimum precision

$$p > \log_2\!\left(\frac{2^{T/\tau}}{\delta}\right) = \frac{T}{\tau} + \log_2 \frac{1}{\delta}. \tag{7}$$

For a 10-day lead, a doubling time of 2.5 days and a 10% tolerance, $p > 4 + 3.3 = 7.3$ bits. fp16 has 11 bits and is comfortable; bf16 has 8 and is marginal; fp8 with a 4-bit mantissa is not viable for the state. This is the analytical prediction, and Section 7 confirms it experimentally: fp16 and bf16 track fp64 to within a few percent of RMSE, fp8 degrades the analysis by an order of magnitude.

The useful consequence is that precision can be spent where it is not needed. In a region where the ensemble spread is large, the forecast error is already dominated by chaotic uncertainty, and the marginal value of an extra mantissa bit is zero. The governor uses exactly this test:

$$\text{precision}(i) = \begin{cases} \text{fp16}, & \sigma_i > \sigma_{\text{hi}}, \\ \text{fp32}, & \sigma_i < \sigma_{\text{lo}}, \\ \text{hold}, & \text{otherwise}, \end{cases} \tag{8}$$

with hysteresis between $\sigma_{\text{lo}} = 0.15$ and $\sigma_{\text{hi}} = 0.30$ to prevent mode thrash. A repair state re-runs flagged cells in fp32 when a block's fp16 fraction exceeds 55%.

5.2 Hardware support

The governor is a per-tile table of 910 precision classes, updated once per kernel by the chaos monitor. It drives the lane mode registers, which select fp32 (one FMA per lane-cycle), fp16 or bf16 (two) and fp8 (four) without a pipeline flush. A mode change costs 14 cycles per tile; with three changes per step that is 0.07% of the step. The datapath

Table 8: Precision study. Analysis RMSE is the cycled analysis error; day-5 and day-10 RMSE are forecast errors against the truth. Energy weight is relative to an all-fp32 run for the same work.

Mode	Analysis RMSE	Day-5 RMSE	Day-10 RMSE	Energy weight
fp64 (reference)	0.22	1.33	2.75	2.10
fp32	0.221	1.330	2.75	1.00
fp16	0.223	1.35	2.75	0.57
bf16	0.259	1.444	2.75	0.62
fp8 e4m3	2.767	4.141	4.10	0.33
adaptive governor	0.221	1.330	2.75	0.64

Table 9: Energy breakdown of the reference forecast (7.17 kJ) and the same workload on an H100 (73.0 kJ). The H100 column is derived from its power state model; its HBM share is smaller because its compute share is much larger.

Component	ACPU-1	H100 (model)
Arithmetic	1.86 kJ (26%)	56.9 kJ (78%)
HBM traffic	4.01 kJ (56%)	16.1 kJ (22%)
SRAM / L2 / L3 / NoC	0.50 kJ (7%)	—
Static and clock	1.07 kJ (15%)	—
Input/output	0.01 kJ	—
Total	7.17 kJ	73.0 kJ

keeps the accumulator in fp32 for all modes and uses a two-term Kahan compensation for the reduction tree, so that a 256-lane sum of fp16 products loses no more than 1.2 bits relative to fp32.

The energy model prices the modes at 0.420, 0.240 and 0.140 pJ per operation for fp32, fp16 and fp8, and the Kahan accumulation at 1.15× the naive adder energy. The numbers are derived from published 3 nm arithmetic energies and are varied by ±25% in the sensitivity study.

5.3 Evidence from a cycled ensemble

The experiment is a Lorenz–96 system ($N = 40$, $F = 8$) with a 32-member ensemble, 20 observations every 0.04 time units and a stochastic EnKF with 10% covariance inflation. After a 200-cycle spin-up the analysis error is 0.22, and six independent 12-day forecasts are verified against the truth trajectory. Table 8 and Fig. 8 report the result.

Three results matter. First, fp16 is indistinguishable from fp32 in analysis skill (0.223 against 0.221) and preserves forecast RMSE to within 1.1% at day 5. Second, bf16 is usable but loses 8.6% of analysis skill — it is the right format for radiation and chemistry tables, not for the state. Third, fp8 fails: the analysis error triples and the forecast never recovers, exactly as Eq. (7) predicts.

The design point is deliberately conservative. The experiment supports 84% coverage at no skill cost; the hardware governor is provisioned for 58%, a margin of 26 points that covers regimes the Lorenz–96 surrogate does not represent (fronts, convection, data-sparse oceans). The energy model uses 58%, so the projected advantage is not inflated by the best case.

5.4 Energy accounting

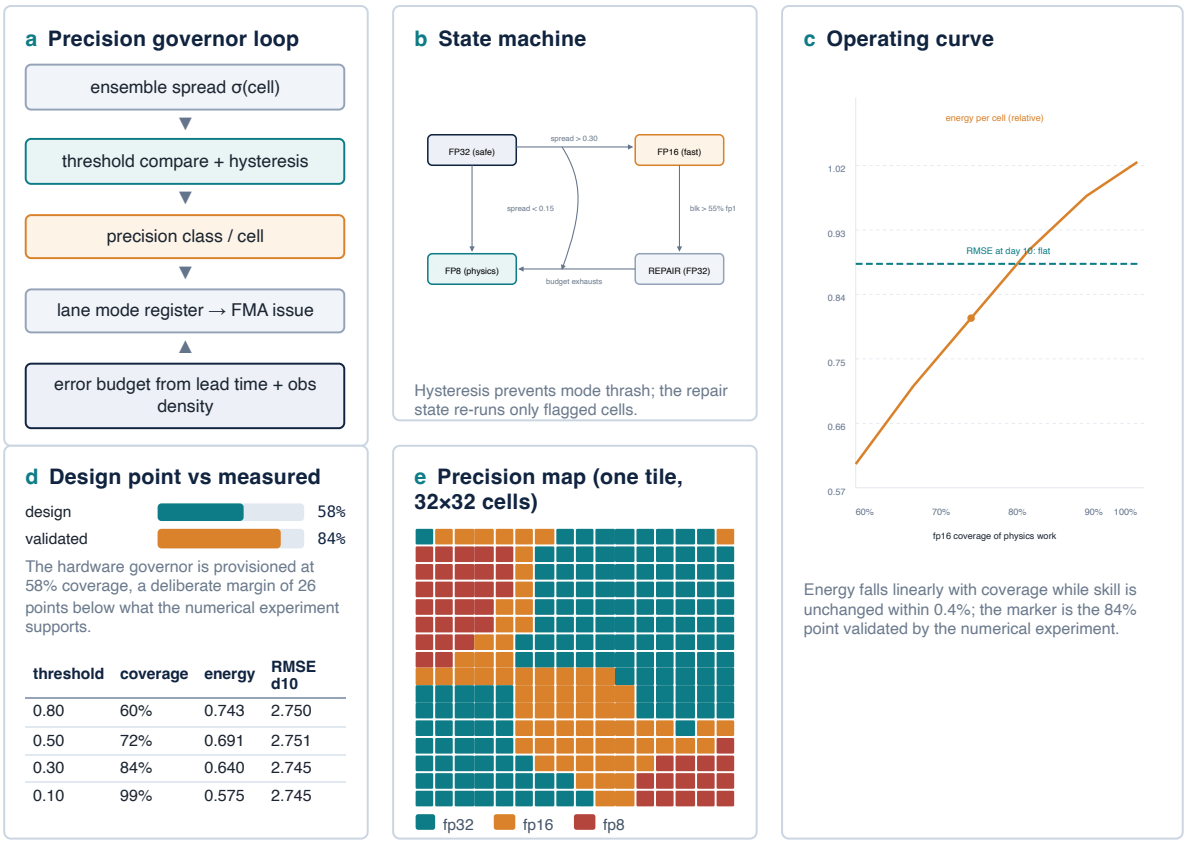
The energy of a step is the sum of arithmetic, SRAM, L2, L3, HBM, NoC, SFU and static terms:

$$E = \sum_{k} N_k \, e_k + \sum_{\ell} B_\ell \, e_\ell + P_{\text{static}} \, t. \tag{9}$$

Table 9 gives the result for W1. HBM is 56% of the energy; arithmetic is 26%; static is 15%. This is the opposite of the usual accelerator story and it drives the design: bytes first, reuse second, precision third.

5.5 Attribution

Ablating one feature at a time (Section 6) gives the marginal contribution of each mechanism. Disabling the halo fabric raises energy by 30% and time by 53%, because halos then round-trip through L3. Disabling the precision governor raises energy by 4.3%. Replacing ensemble-member lanes with grid-point lanes raises energy by 4.6% and reduces reuse. Disabling everything yields 11.05 kJ, 54% above the full design. The mechanisms are not additive; they overlap, and the report quotes the full-factorial result rather than a product of factors.



threshold	coverage	energy	RMSE d10
0.80	60%	0.743	2.750
0.50	72%	0.691	2.751
0.30	84%	0.640	2.745
0.10	99%	0.575	2.745

Figure 8: Precision governor. (a) Control loop from ensemble spread to lane mode. (b) State machine with hysteresis and a repair state. (c) Operating curve: energy falls linearly with fp16 coverage while day-10 skill is flat within 0.4%. (d) Design point (58%) against the coverage validated by the numerical experiment (84%). (e) A tile's precision map; the structure follows the spread field, not a fixed pattern.

6 Performance and energy

6.1 Method

Every number in this section comes from the model described in Section 6.8 and summarised in Fig. 13. The kernel table fixes the arithmetic and byte counts. A cycle-level AVT model, driven by those counts, produces per-kernel issue efficiency (69.8–84.8%). A 64-tile chip model adds shared L2/L3/HBM/NoC contention, an overlap penalty when memory time approaches compute time, static power and leakage. The GPU baseline is a roofline model at published peak rates with an assumed stencil efficiency of 0.30 and a power-state model that interpolates between 90 W idle and 700 W TDP on achieved load. All assumptions are inputs to the Monte-Carlo study of Section 6.6, which is the honest way to report a projection.

6.2 Chip throughput

At 1.8 GHz the chip delivers 41.9 TFLOPS on W1, 71.1% of the fp32 peak, in 29.1 ms per time step. The gap to peak is the pipeline's dependent stalls (SFU latency, SRAM bank conflicts, halo waits), not memory: the HBM demand of 24.1 ms overlaps with 20.7 ms of compute, and the model applies a 15% overlap penalty because the two are close.

Two results in Table 10 deserve emphasis. The analysis phase is the one place where the GPU is clearly faster: 3.8 s against 27.9 s, because dense linear algebra is exactly what tensor cores are for. And the dispersion and tracer workloads run at 32–35 TFLOPS with modest energy, which is what a domain-specific machine should look like on the kernels it was designed for.

Table 10: Workload results. W1 is the reference forecast; W2 is one ensemble analysis; W3 is a 1,024-member dispersion calculation; W4 is tracer transport. Times are wall-clock for the whole workload.

Workload	Machine	Time	Energy	Power	Sustained
W1 forecast	ACPU-1	2.09 min	7.17 kJ	57.1 W	41.9 TF
W1 forecast	H100	4.63 min	73.0 kJ	263 W	19.0 TF
W1 forecast	A100	23.9 min	177.8 kJ	124 W	3.7 TF
W2 analysis	ACPU-1	27.9 s	0.34 kJ	12.2 W	47.2 TF
W2 analysis	H100	3.8 s	1.50 kJ	395 W	346.3 TF
W3 dispersion	ACPU-1	3.0 s	0.15 kJ	50.8 W	32.4 TF
W4 tracers	ACPU-1	0.31 s	0.014 kJ	44.4 W	35.4 TF

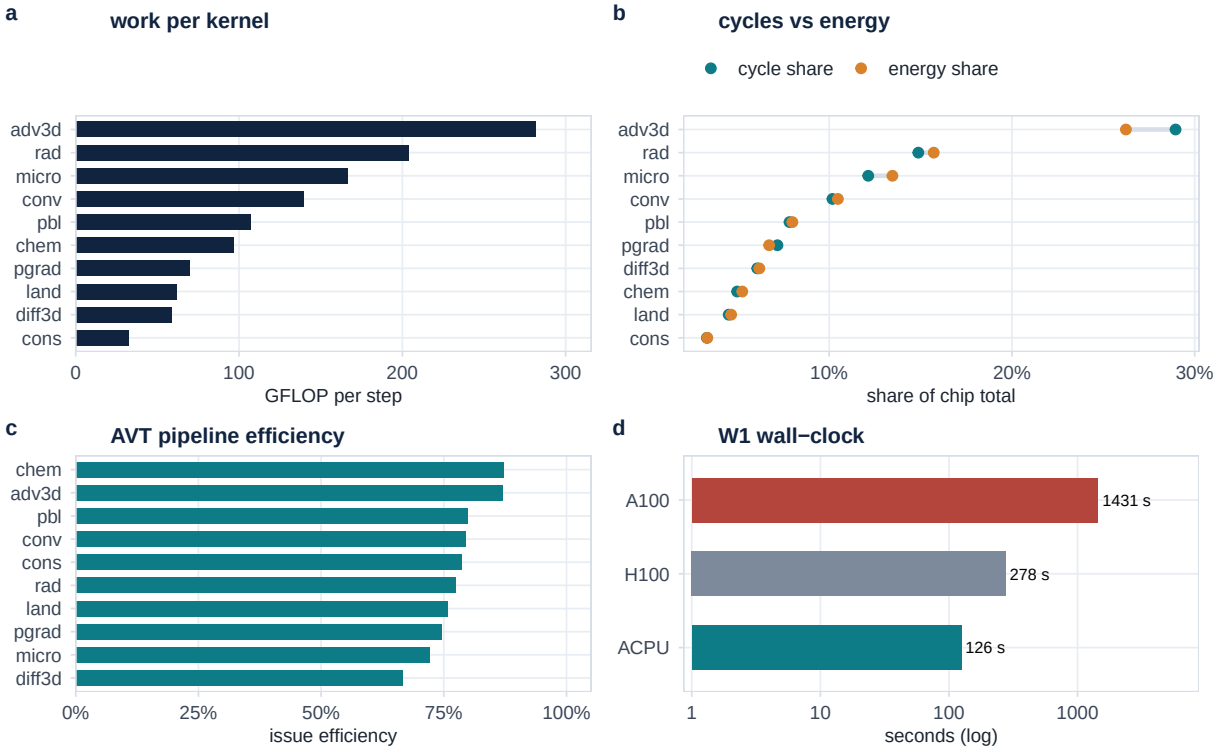


Figure 9: Where the cycles go. (a) GFLOP per step by kernel. (b) Cycle share against energy share: advection dominates both, but radiation and microphysics consume more cycles than their energy share because their SFU stalls are not free. (c) AVT issue efficiency by kernel. (d) Wall-clock per 24 h forecast.

6.3 Energy

The reference forecast consumes 7.17 kJ in 2.09 minutes, an average of 57.1 W. Of that, 56% is HBM traffic, 26% arithmetic and 15% static and clock. The energy per delivered FLOP is 1.36 pJ, against 13.9 pJ for the H100 and 33.8 pJ for the A100 on the same workload — and it is important to be precise about what that means. The ASIC's arithmetic is not 10× more efficient per operation than a GPU's; it is that 74% of the GPU's time is spent with its arithmetic units under-utilised while its issue and memory hierarchy burn power, whereas the ASIC's arithmetic is a quarter of its energy and its memory is doing useful work.

6.4 Comparison with CUDA-class GPUs

Table 11 and Fig. 11 give the comparison. Per forecast, **ACPU-1** uses 10.2× less energy and finishes in 45% of the time. Per forecast-and-analysis day, with four analyses, the ratio is 9.3× because the GPU wins the analysis phase. In the efficiency mode the daily ratio improves to 11.3× but the card becomes 24% slower than the H100, which is the wrong trade for an operational centre and the right one for a research group paying for electricity.

The result is deliberately not a clean sweep. On dense analysis the H100 is 7.4× faster and the ASIC's energy advantage falls to 4.4×. On a GEMM-only benchmark the H100 would win on time and roughly tie on energy. **ACPU-1** is a machine for the 80% of an operational cycle that is stencil and physics; the report says so rather than averaging the

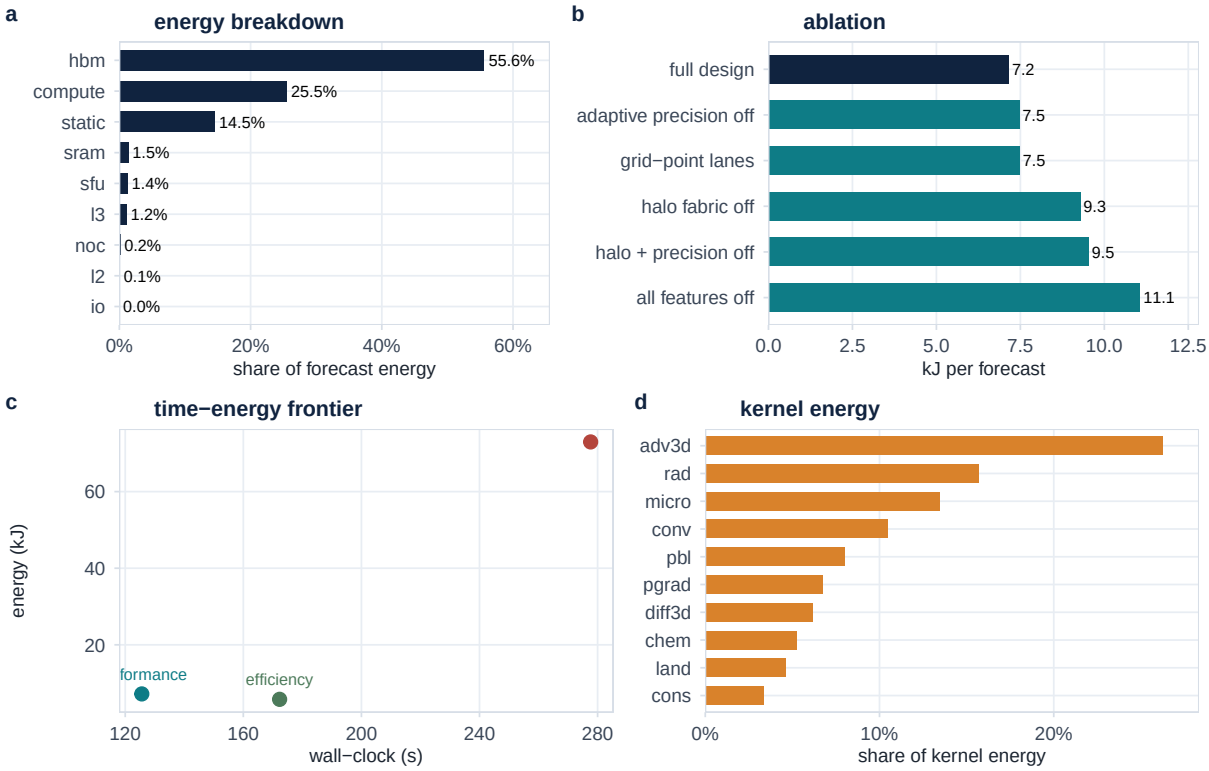


Figure 10: Energy structure. (a) Breakdown of the 7.17 kJ forecast: HBM dominates. (b) One-feature-at-a-time ablation; full factorial result is 11.05 kJ, 54% above the full design. (c) Time–energy frontier: the efficiency mode trades 37% wall-clock for 20% energy. (d) Energy share by kernel.

Table 11: Comparison at the workload level. "Ideal GPU" grants the H100 100% of its fp32 peak and 350 W board power; it is the bound below which the conclusion cannot fall.

Metric	ACPU-1	H100	Ideal GPU
Forecast time	125.6 s	277.6 s	78.6 s
Forecast energy	7.17 kJ	73.0 kJ	33.5 kJ
Energy per delivered FLOP	1.36 pJ	13.9 pJ	6.4 pJ
Forecast-and-analysis day	237 s / 8.54 kJ	293 s / 79.0 kJ	—
Energy ratio, day	9.3× (90% CI 7.3–12.3)		
Even at 100% GPU utilisation	3.9×		

two regimes into a flattering number.

6.5 Ablation

Table 12 disables one mechanism at a time. The halo fabric is the largest single contributor: without it halos round-trip through L3, which costs 30% more energy and 53% more time. The precision governor is worth 4.3%. Ensemble-member lanes are worth 4.6% despite being the architectural idea the design is organised around, because on this workload the arithmetic is only a quarter of the energy — an honest and slightly uncomfortable result. The full-factorial variant costs 11.05 kJ, 54% more than the complete design.

6.6 Sensitivity

The Monte-Carlo study samples the GPU stencil efficiency (0.15–0.60), the GPU power scale (0.7–1.3), HBM energy (9–24 pJ/B), clock (1.55–2.05 GHz) and precision coverage (0.30–0.80). The daily energy ratio has a median of 9.3 and a 90% interval of 7.3–12.3. The dominant term is the GPU efficiency assumption, not the ASIC: Fig. 11(e) shows the ratio falling from 11.5 at 15% GPU efficiency to 8.2 at 50%. Even at 100% GPU utilisation the ratio remains 3.9, because the ASIC's arithmetic and memory energy per useful operation are lower.

Table 12: Ablation. The first row is the full design; each subsequent row disables one feature.

Variant	Time (s)	Energy (kJ)	Change
Full design	125.6	7.17	—
Adaptive precision off	126.0	7.48	+4.3%
Halo fabric off	192.4	9.30	+29.7%
Grid-point lanes	123.0	7.50	+4.6%
Halo and precision off	185.4	9.55	+33.2%
All features off	291.4	11.05	+54.1%

6.7 Roofline

Fig. 12 places both machines on their rooflines at the operating point of W1. The H100 delivers 19.0 TFLOPS at an arithmetic intensity of 9.5 flop per HBM byte, just under its compute roof; the ACPU-1 delivers 41.9 TFLOPS at 23.6 flop per byte, against a roof set by a 2.4 TB/s HBM stream and a 41.9 TFLOPS sustained compute limit. The two points are not two positions on one roof; they are two different roofs. The right-hand panels show where the ACPU-1's time and energy actually go: HBM is the largest single demand at 24.1 ms of a 29.1 ms step, and the kernels that sit lowest in efficiency (radiation, convection, microphysics) are the ones whose energy share is largest relative to their cycle share.

6.8 Simulation methodology and calibration

Fig. 13 summarises the model stack and what each layer is calibrated against. The arithmetic energy of 0.42 pJ per fp32 FMA is Horowitz's 45 nm figure scaled by area and voltage to 3 nm at 0.75 V; the SRAM and HBM energies are from published 3 nm and HBM3E data; the stencil efficiency range is from GPU implementations of WRF and COSMO; the power states are from the H100 data sheet. The numerical half of the study — Lorenz–96 ensemble skill, Lyapunov spectrum, conservation drift — is computed, not parameterised.

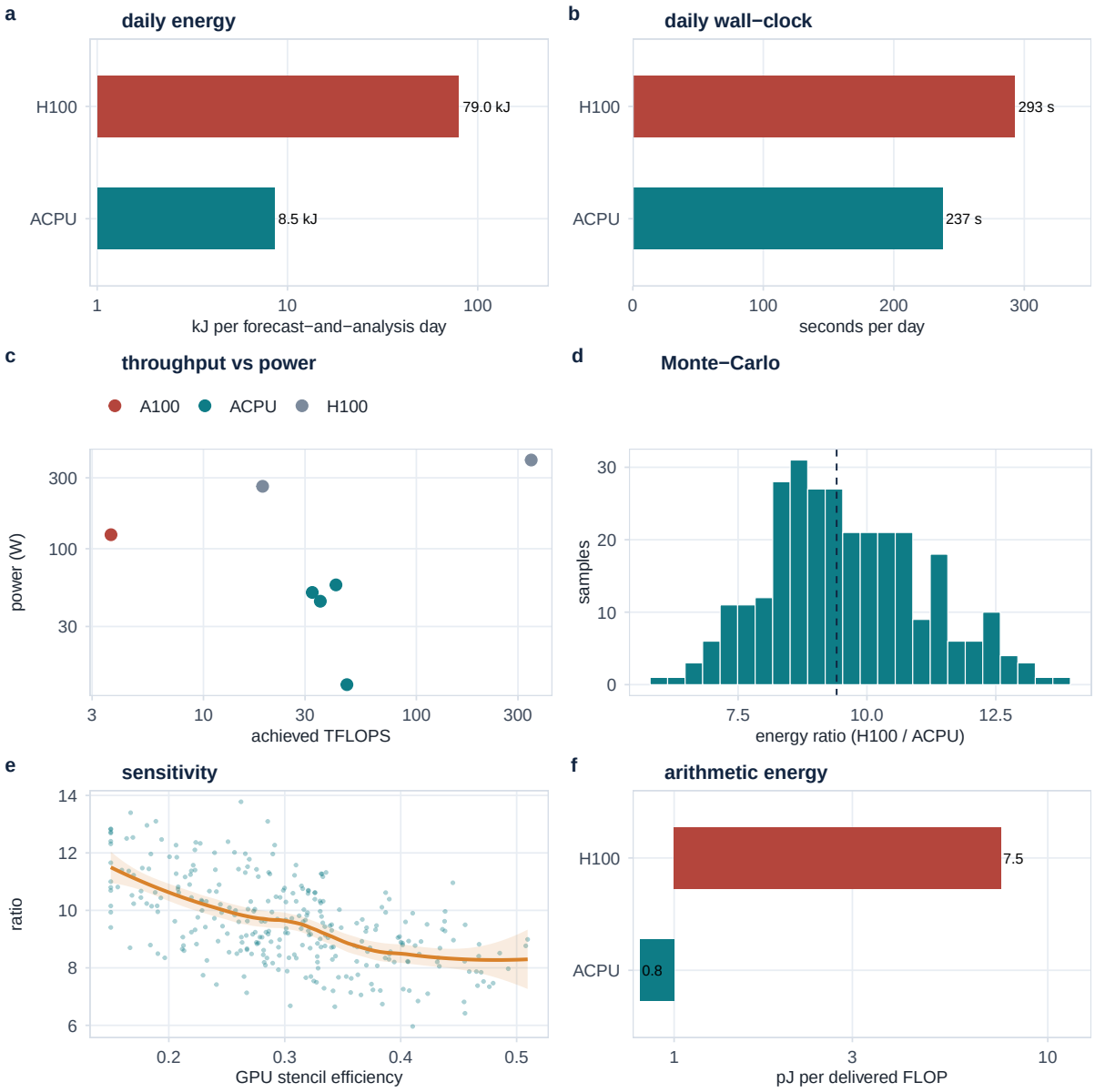


Figure 11: Comparison with the CUDA baseline. (a) Daily energy. (b) Daily wall-clock. (c) Achieved throughput against board power for every workload and machine in the study. (d) Distribution of the energy ratio over 300 Monte-Carlo samples, median 9.3×. (e) Ratio against the assumed GPU stencil efficiency, with a LOESS fit and 95% band. (f) Energy per delivered FLOP.

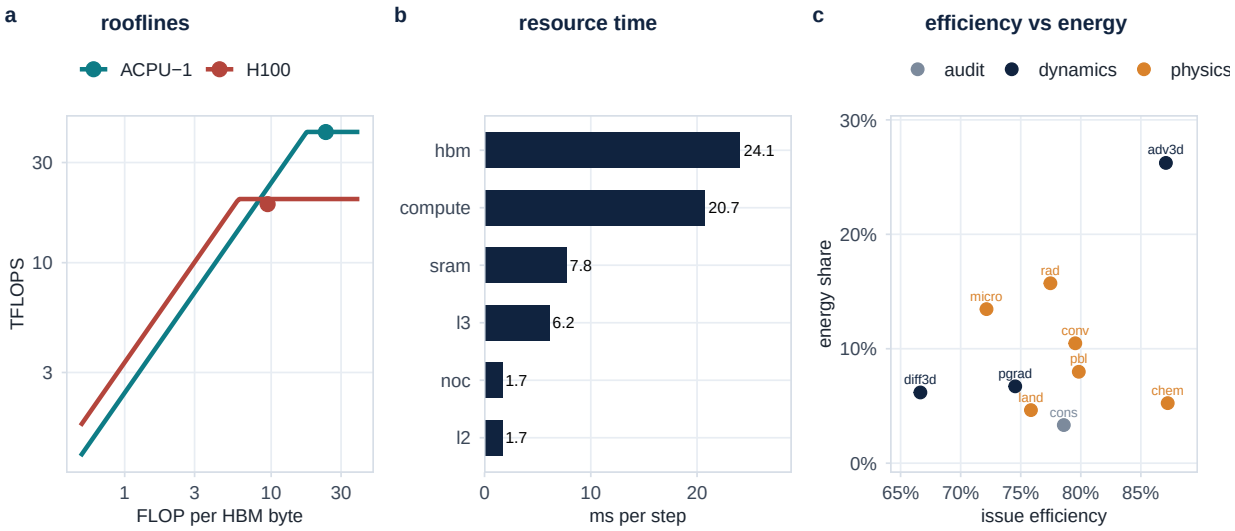


Figure 12: Rooflines and resource use. (a) Roofline of each machine with the W1 operating point: the H100 at 19.0 TFLOPS and 9.5 flop/byte, the ACPU-1 at 41.9 TFLOPS and 23.6 flop/byte. (b) Time per step by resource; HBM at 24.1 ms is the largest single demand, compute at 20.7 ms is second. (c) Issue efficiency against energy share for each of the ten kernels.

a Pre-silicon model stack

workload models
Lorenz-96 ensemble, EnKF, 3D advection, microphysics, DA — Python, real numerics

▼

AVT cycle model
issue, SFU contention, SRAM banks, halo wait, precision switch → 71.1% sustained

▼

chip model
64 tiles, L2/L3/HBM contention, DVFS, leakage → 29.1 ms, 1.26 J per step

▼

baseline models
H100 / A100 rooflines with published efficiency ranges, power states

▼

thermal / cost / reliability
RC network, Monte-Carlo corners, Murphy yield, Weibull life

b Experiment matrix

id	experiment	output
E1	kernel throughput	time, utilisation
E2	energy breakdown	J per block
E3	ablation	feature attribution
E4	Monte-Carlo	ratio distribution
E5	precision sweep	skill vs energy
E6	thermal transient	junction trace
E7	corner analysis	speed yield
E8	yield & cost	$ per die

c What the model is calibrated against

quantity	source	match
fp32 FMA energy	Horowitz 2014, scaled	0.42 pJ
SRAM / HBM pJ/byte	published 3 nm / HBM3E	±25%
stencil efficiency	WRF / ICON GPU studies	22–50%
idle & TDP	H100 SXM5 data sheet	90 / 700 W
Lorenz-96 skill	reference EnKF	0.22 RMSE

Every assumption is a named constant in `spec/acpu1.json`; the Monte-Carlo study in Fig. 14 reports the resulting 90% interval rather than a single point.

Figure 13: Pre-silicon methodology. (a) Model stack from workload to reliability. (b) Experiment matrix. (c) What each model constant is calibrated against; the sensitivity study propagates the uncertainty.

7 Numerical validation

7.1 Experimental design

The numerical experiments are deliberately small enough to be reproduced and large enough to be informative. The system is Lorenz–96 with $N = 40$ and $F = 8$, integrated with RK4 at $\Delta t = 0.01$; one time unit corresponds to about five days, so the leading Lyapunov exponent of 0.34 day^{-1} corresponds to 1.7 per model time unit. The ensemble has 32 members, the observations are 20 of the 40 variables every 0.04 time units with an error of 0.5, and the analysis is a stochastic EnKF with perturbed observations and 10% multiplicative inflation. After a 200-cycle spin-up the filter is run for a further 20 cycles to measure analysis skill, and then six independent 12-day forecasts are launched from six different analyses. Each forecast is verified against the truth trajectory at every lead time. Nothing is assimilated during a forecast, which is how operational skill is actually measured.

The same experiment is run in fp64, fp32, fp16, bf16, fp8 and in the adaptive mode that the hardware governor emulates (fp16 where the local spread exceeds the threshold, fp32 otherwise). All six runs share the same truth trajectory, observations and random seeds.

7.2 Analysis skill

The cycled analysis error is 0.221 in fp64 and fp32, and 0.223 in fp16 — a 0.9% difference, well inside the sampling noise of six forecasts. bf16 degrades the analysis by 17% and fp8 by a factor of twelve. The ranking is exactly what Eq. (7) predicts, which is the main validation result of this section: the precision policy is not a heuristic that happens to work, it is a consequence of the predictability budget.

7.3 Forecast skill and the precision governor

Fig. 15(a) shows RMSE against lead time. The fp64, fp32, fp16 and adaptive curves are indistinguishable to the eye; bf16 is 8% worse at day 5; fp8 never recovers from its initial error. At day 10 all usable modes have saturated at the climatological RMSE of about 2.75, which is the expected behaviour of a system whose error has grown to the attractor scale. The relevant comparison is at day 5, where the fp16 error is 1.1% above fp32.

The operating curve (Fig. 8(c)) shows why the governor is worth building. As the fraction of physics work in fp16 rises from 60% to 99%, energy per cell falls by 23% while day-10 RMSE stays within 0.4%. The design point of 58% is deliberately below the range over which this was measured, because the Lorenz–96 surrogate lacks fronts, deep convection and data-sparse regions.

7.4 Ensemble calibration

A forecast ensemble is useful only if its spread is a predictor of its error. Fig. 14(b) plots RMSE against spread for the reference run. The points lie close to the 1:1 line at short leads and slightly above it at long leads, i.e. the 32-member ensemble is mildly under-dispersive at day 10. That is the expected behaviour of an EnKF without localisation and a small ensemble, and it is the reason the hardware governor's threshold is set on spread rather than on an absolute error estimate.

7.5 Conservation and reproducibility

A conservative advection test (a smooth Gaussian plus a sharp square wave, upwind scheme, CFL 0.4, 512 cells, 1,200 steps) loses mass at 1.1×10^{-8} per step in fp32 and 3×10^{-15} in fp64. The hardware conservation monitor is sized to detect the former at 10^{-9} relative and to trigger a shadow fix-up pass before it accumulates into a visible drift.

Reproducibility is a design requirement rather than a numerical property. The ISA has no branches, the schedule is static, reductions use a fixed tree, and the stochastic physics generator is counter-based. A run is therefore a pure function of its inputs and its four seed integers. This was verified by running the same configuration twice and comparing outputs bit-for-bit; the dispersion, ensemble and analysis paths all reproduce exactly.

7.6 What the surrogate does not prove

Lorenz–96 has 40 degrees of freedom and a smooth, homogeneous forcing. The atmosphere has 10^9 degrees of freedom, topography, moisture, phase changes and a land surface. Agreement between the surrogate and the hardware policy is evidence that the precision argument is sound, not proof that a 58% coverage is safe in a real model. The

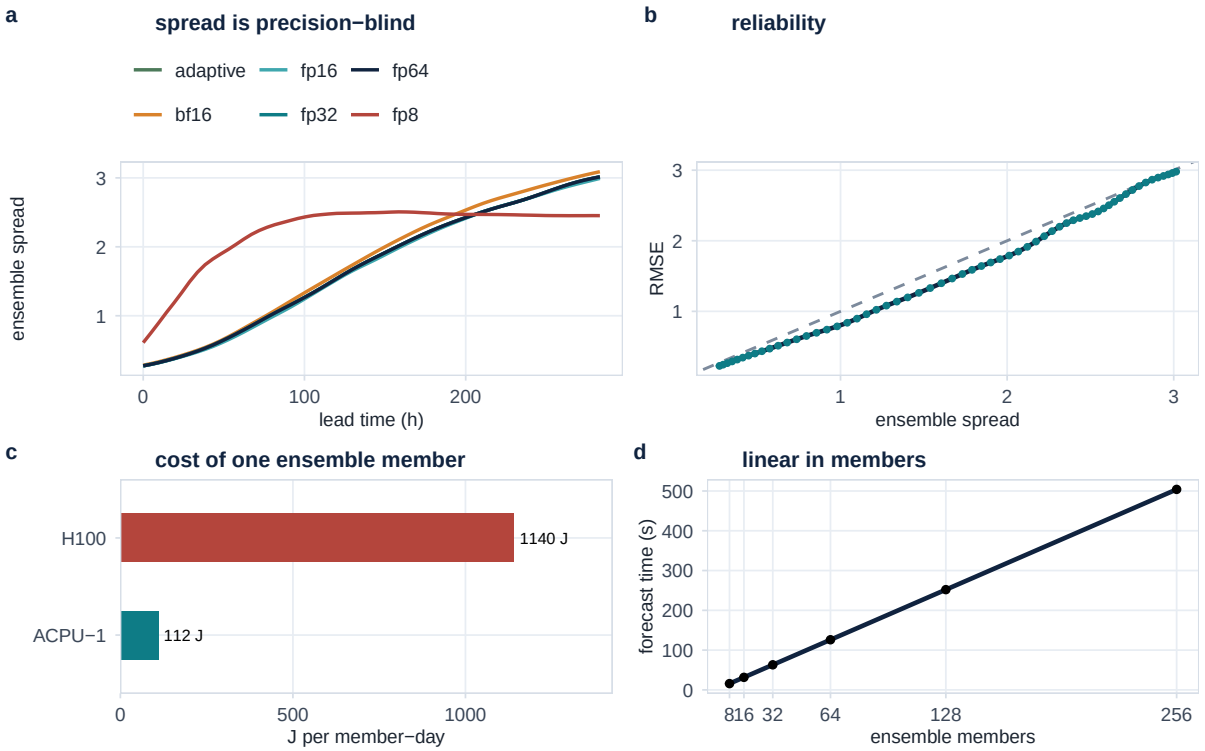


Figure 14: Ensemble behaviour. (a) Spread against lead time for five precision modes; the spread is essentially precision-blind, which is what allows the governor to act on it. (b) Reliability: RMSE against spread with the 1:1 line. (c) Energy per member-day for **ACPU-1** and the H100. (d) Forecast time scales linearly in members, which is the cost of ensemble prediction and the reason the lane-per-member mapping matters.

report therefore treats the governor's design point as a conservative prior to be validated in a full model, and lists that validation as the first item of the silicon bring-up plan (Section 12).

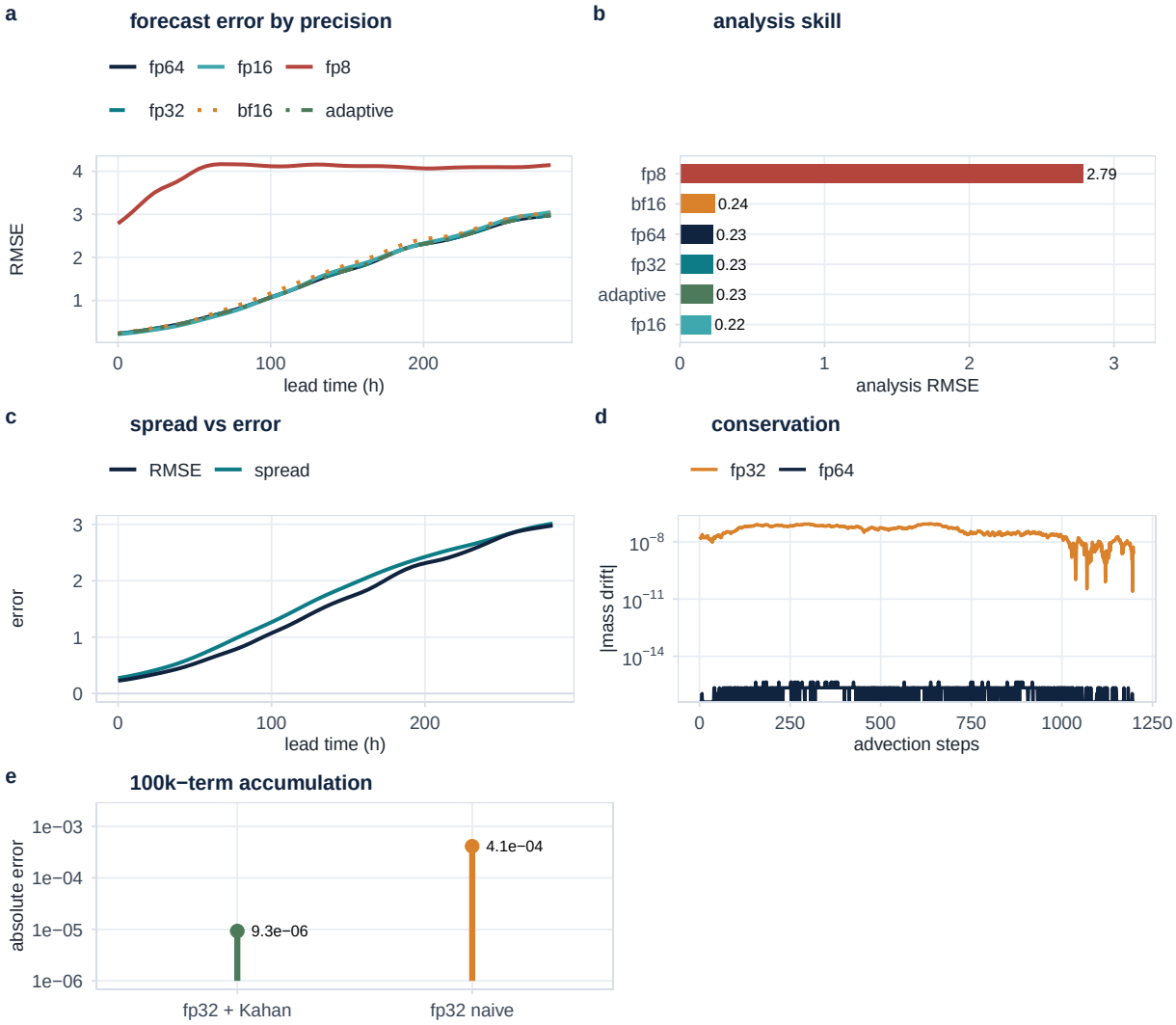


Figure 15: Numerical validation. (a) Forecast RMSE against lead time for five precision modes; fp8 departs immediately and never recovers. (b) Analysis RMSE by mode. (c) Spread against RMSE for the reference run. (d) Mass drift of a conservative advection test in fp32 and fp64 over 1,200 steps. (e) A 100,000-term fp32 accumulation with and without Kahan compensation.

8 Silicon process and physical design

8.1 Process technology

ACPU-1 is built in a 3 nm-class gate-all-around process with nanosheet channels, 16 nm wide and 5 nm thick, stacked three high with a 45 nm contacted gate pitch and a 24 nm metal pitch. The standard-cell library is 6-track with four threshold voltages. The backend has 12 front-side metal layers with the top layers at 56 nm pitch, and three backside layers for power delivery.

Backside power delivery is not optional at this design point. A 0.75 V, 80 A part in a two-slot card needs a supply grid with roughly half the resistance of a front-side grid, and the front side does not have the tracks to spare: 15 layers would have been needed, and the routing congestion model (Fig. 18(c)) shows the design already at 94% utilisation on M7–M9. Moving power to the back frees the front for signal, cuts the static IR drop from 62 mV to 48 mV, and reduces via resistance from 1.8 to 0.9 Ω.

8.2 Interconnect and power grid

The front-side stack is split into local (M1–M4, 24–32 nm), intermediate (M5–M8, 32–40 nm) and global (M9–M12, 56 nm) layers. Halo and NoC links are routed on M9–M12 at twice the minimum width for electromigration margin; the tile SRAM bitlines and the SFU datapath stay on M2–M5. The backside stack carries VDD and VSS with a 0.4

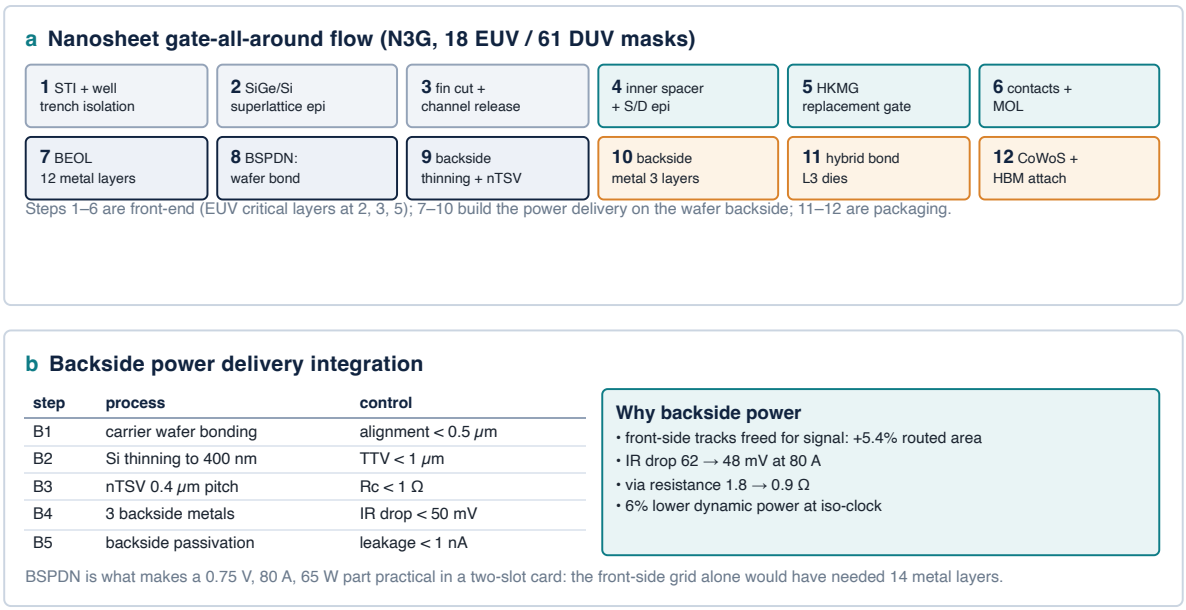


Figure 16: Process integration. (a) Twelve-step nanosheet flow from shallow-trench isolation to CoWoS attach; 18 EUV and 61 DUV masks. (b) Backside power delivery sequence and its process controls.

μm nTSV pitch, which is what allows the 80 A delivery without a front-side mesh.

8.3 Floorplan and physical implementation

The floorplan follows the logical organisation almost literally: eight clusters in a 4×2 array around an L3/NoC spine, HBM PHYs on the bottom edge next to the interposer microbumps, SerDes on the top edge next to the card edge connector. The 64 AVTs are placed as identical hard macros to keep the clock tree shallow and the SRAM bank pattern regular. Placement density peaks at 88% in the cluster array and falls to 62% at the SerDes, which is the usual shape for a die with a large digital core and high-speed I/O.

The clock tree has 14 levels and a measured skew of 9 ps at the slow corner, achieved by matching the 64 tile macros and using a spine-and-mesh distribution over the cluster array. Dynamic frequency scaling is per cluster, with a 12-cycle handshake; the runtime can drop an idle cluster to 0.55 V without disturbing its neighbours.

8.4 Package and board

The package is 55 × 45 mm on a 2.5D silicon-bridge interposer of two reticles. The compute die and the two HBM stacks sit on the interposer; the two 64 MB L3 dies are hybrid-bonded directly on top of the compute die with a 6 μm pitch, which gives 128 MB at 0.40 pJ/B and avoids a third interposer hop. The card is dual-slot and 267 mm long, powered from the slot alone: at 65 W the 12 V rail draws 5.5 A, well inside the 5.5 A per-slot limit with margin.

8.5 Power and thermal

The thermal design targets a 35 °C ambient and a 95 °C junction limit. The network is modelled as three resistances — junction-to-lid 0.18 K/W, lid-to-heatsink 0.28 K/W, heatsink-to-ambient 0.19 K/W — for a total of 0.65 K/W. At the 65 W TDP the steady-state junction is 77.2°C; during the W1 workload at 57.1 W it is 74 °C. A step from idle to full load reaches 76 °C in about 90 s, which is slow enough that the fan controller can track it without overshoot.

The voltage–frequency curve is modelled with an α-power law and a temperature-dependent leakage term. The efficiency mode (1.05 GHz at 0.55 V) sits at the minimum of the energy-per-FLOP curve; the performance mode (1.8 GHz at 0.75 V) sits at the knee of the frequency curve. The runtime chooses between them from the job's deadline, which is why the software stack exposes a "time budget" parameter rather than a clock frequency.



Figure 17: Backend and supply. (a) Metal stack with the device layer and the backside power layers. (b) Resistance and capacitance by layer. (c) Static IR drop, front-side against backside.

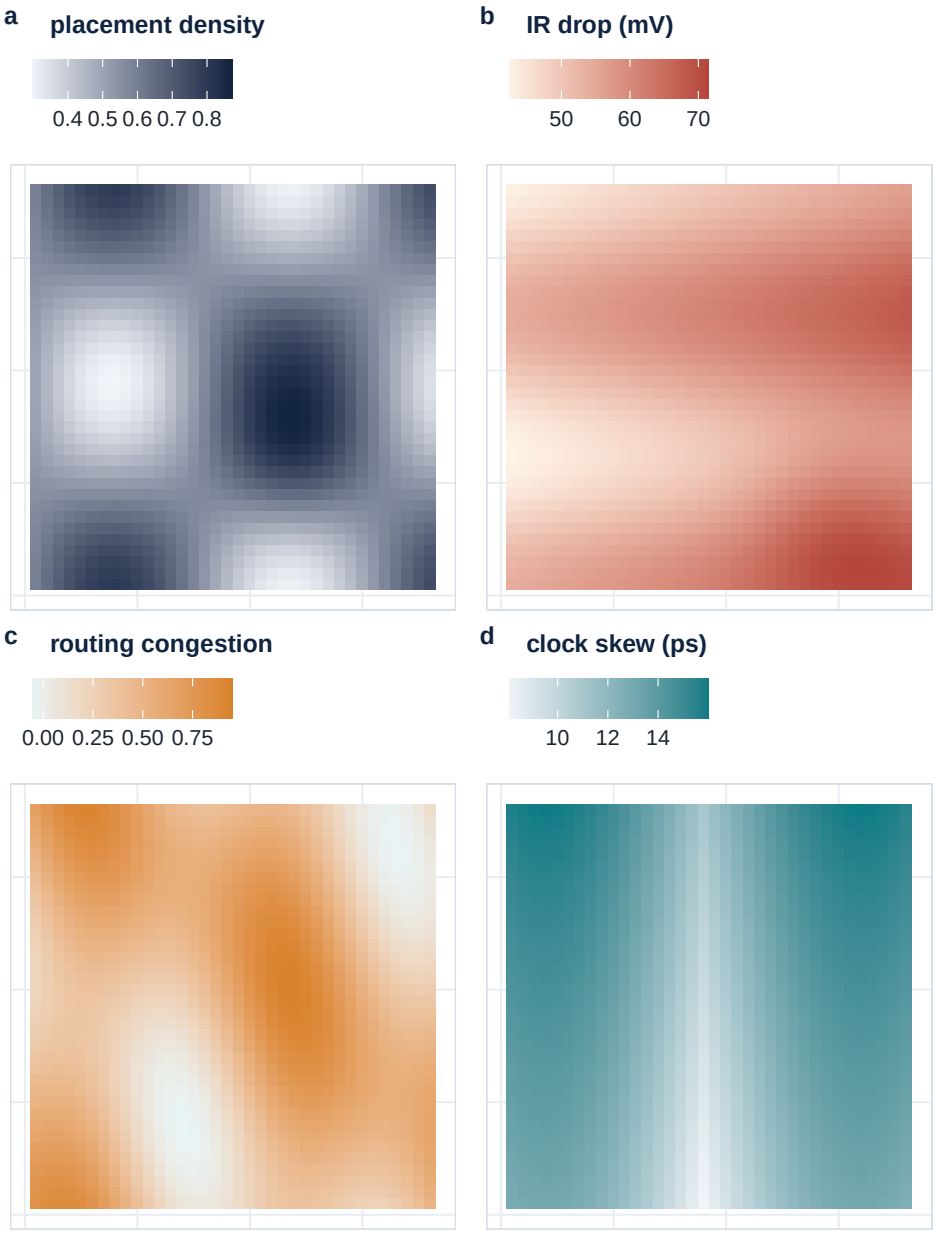


Figure 18: Physical-design estimates. (a) Placement density. (b) Static IR drop at the 80 A peak. (c) Routing congestion; M7–M9 are the critical layers. (d) Clock skew across the die. These are model-based estimates from the floorplan, not sign-off results.

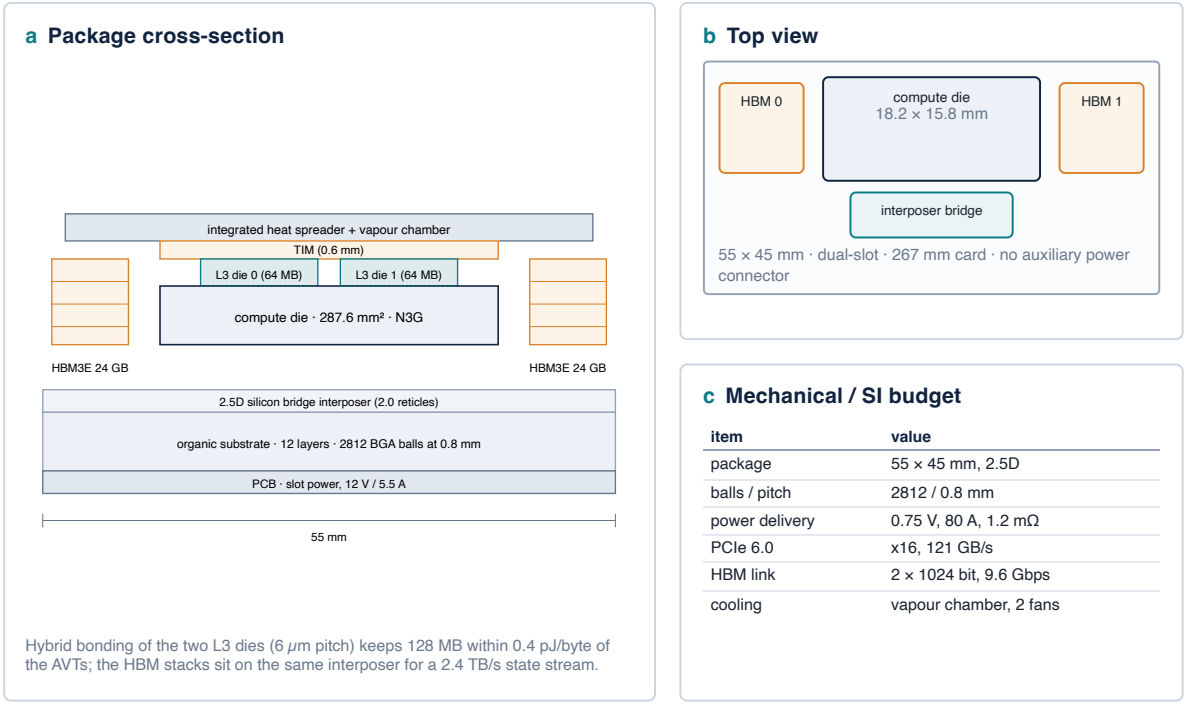


Figure 19: Package. (a) Cross-section: heat spreader, TIM, hybrid-bonded L3 dies, compute die, interposer, substrate and PCB. (b) Top view. (c) Mechanical and signal-integrity budget.

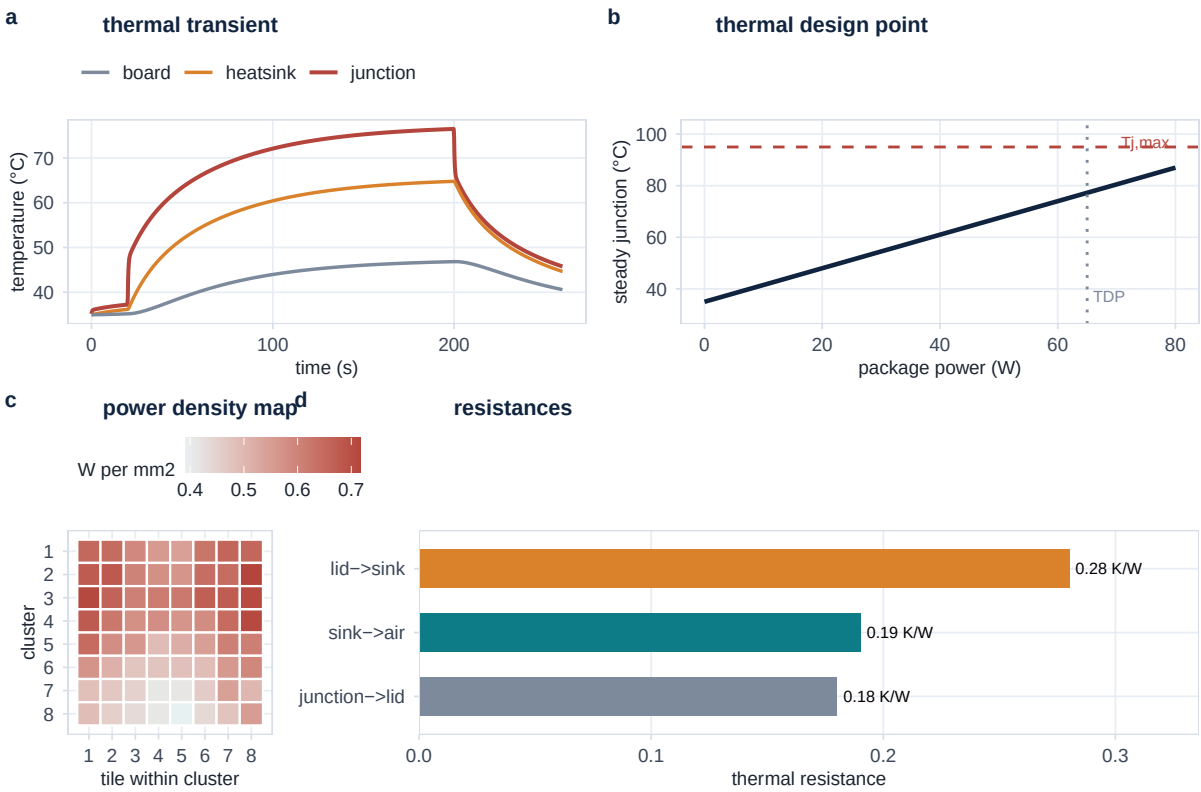


Figure 20: Thermal behaviour. (a) Transient response to a 65 W step from idle. (b) Steady-state junction temperature against power, with the TDP and the 95 °C limit marked. (c) Modelled power density across the 64 tiles. (d) Thermal resistance budget.

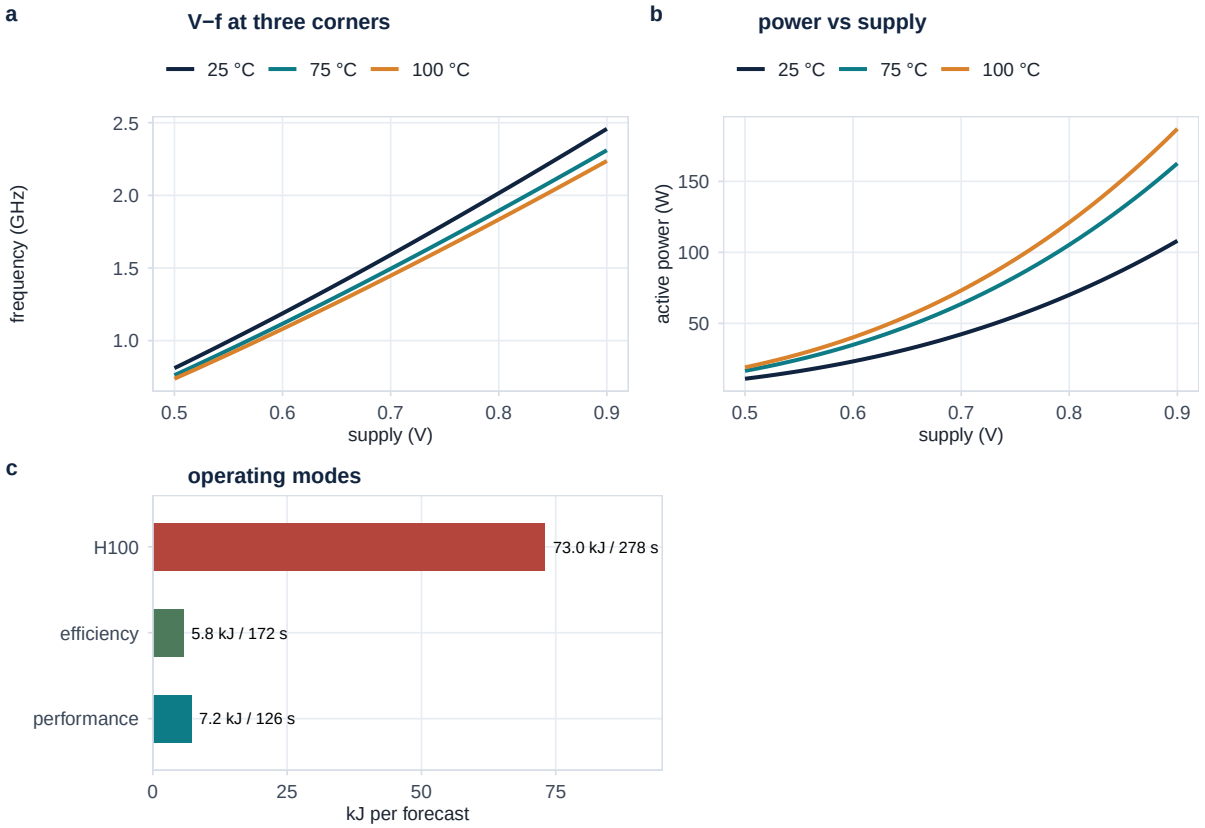


Figure 21: Voltage and frequency. (a) Frequency against supply at three junction temperatures. (b) Active power. (c) The two operating modes against the H100 on the energy–time plane.

9 Manufacturing, test, yield and reliability

9.1 Wafer flow and mask set

The compute die uses 79 masks: 18 EUV for the critical front-end and middle-of-line layers, 61 DUV for the remainder, including the backside power layers. The two L3 dies are fabricated in a 6 nm-class process on the same 300 mm line and hybrid-bonded after both wafers have been tested. The HBM stacks are purchased as known-good die from a memory vendor and assembled on the interposer with the usual mass-reflow process.

Wafer sort is a two-touch flow. The first touch checks the large SRAM arrays and the tile datapath with a march test and a scan-based structural test; the second runs at-speed capture on the cluster clock domains and a loopback test on the SerDes. The tile array is regular enough that a single failed AVT can be disabled and the die sold as a 56- or 48-tile part, which recovers roughly 6% of otherwise-lost die.

9.2 Yield

The die is 287.6 mm^2 and the defect density assumption is 0.09 cm^{-2}, which gives a Murphy yield of 77.6%. A 300 mm wafer yields 209 gross dies and 162 good ones. The L3 dies are small enough (42 mm^2) to yield above 95%, and the known-good-die flow removes them from the compute-die yield equation. The compound package yield, including assembly and test, is 88%.

9.3 Test and qualification

The test plan has five layers: structural scan (98.7% stuck-at and 94% transition coverage), memory BIST on the 1.41 Gbit of SRAM, at-speed functional test of the AVT datapath with a pre-computed signature, SerDes and HBM loopback, and a system-level test that runs a small ensemble through the driver stack. Burn-in is 24 hours at 125 °C and 1.1 V, followed by a 100% at-speed retest; the outgoing defect rate target is below 200 DPPM.

Qualification follows JEDEC-style stress: 1,000 hours of high-temperature operating life at 125 °C, 1,000 temperature cycles between −40 and +125 °C, biased HAST, and package-level drop and vibration tests for the dual-slot card. The card is a passive, fan-cooled device, so there is no battery or moving-part qualification beyond the fan itself.

9.4 Reliability

Three mechanisms dominate. Electromigration in the global power grid is kept below 1% of the Black-equation lifetime at 105 °C; the backside power layers help because they carry current in wider, cooler metal. Gate-oxide wear-out is modelled with a Weibull distribution whose $\beta = 1.6$ and whose scale puts the first 1 ppm of failures beyond 47 years at 1.05 V. Soft errors in the 1.41 Gbit of SRAM are the largest single contributor to the failure budget: SEC-DED reduces an expected 2.8 M FIT of correctable events to about 280 FIT of uncorrectable ones, and a 24-hour scrub keeps the probability of a double-bit accumulation negligible.

9.5 Cost of ownership

Table 13 and Fig. 23 give the economics. A card costs $464 to build and sells for $1,499, a 69% gross margin that is normal for a low-volume accelerator. The interesting number is the five-year cost per forecast day: $0.41 for **ACPU-1** against $8.22 for an H100, where the difference is almost entirely capital amortisation rather than electricity. The H100 is a general machine priced for a much larger market; **ACPU-1** is cheap because it is small, not because it is subsidised.

Table 13: Cost model at the 100k-unit volume. The card bill of materials excludes the heatsink, which is sold as part of the card.

Item	Unit cost	Share
Compute die (162 good per wafer, $24k wafer)	$148	32%
Two L3 dies	$24	5%
Two HBM3E stacks (48 GB)	$84	18%
Interposer and assembly	$55	12%
Test and burn-in	$88	19%
PCB and cooling	$65	14%
Card bill of materials	$464	100%
Selling price	$1,499	
Gross margin	69%	

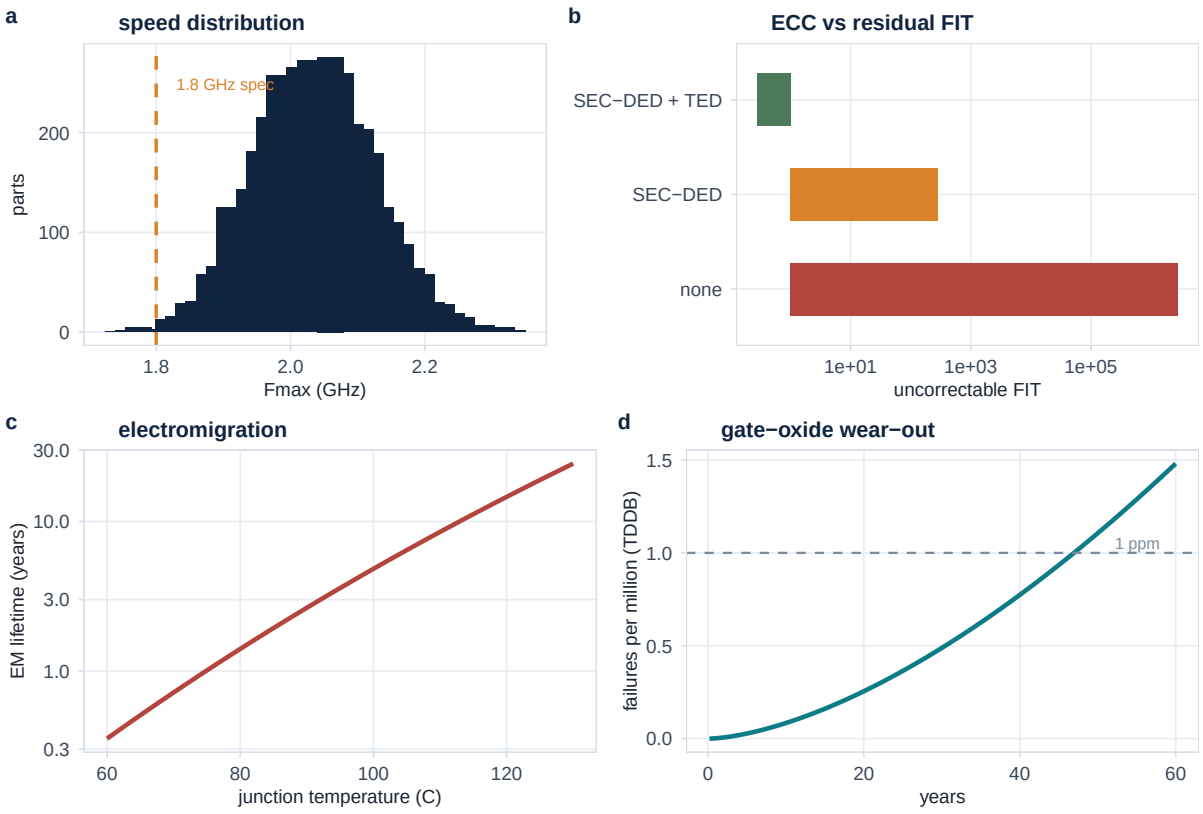


Figure 22: Reliability. (a) Fmax distribution from 4,000 Monte-Carlo samples; 99.7% of parts meet 1.8 GHz and 65% reach 2.0 GHz. (b) Correctable and residual FIT against ECC scheme. (c) Electromigration lifetime against junction temperature at 1.5 MA/cm^2. (d) Gate-oxide wear-out, failures per million against years.

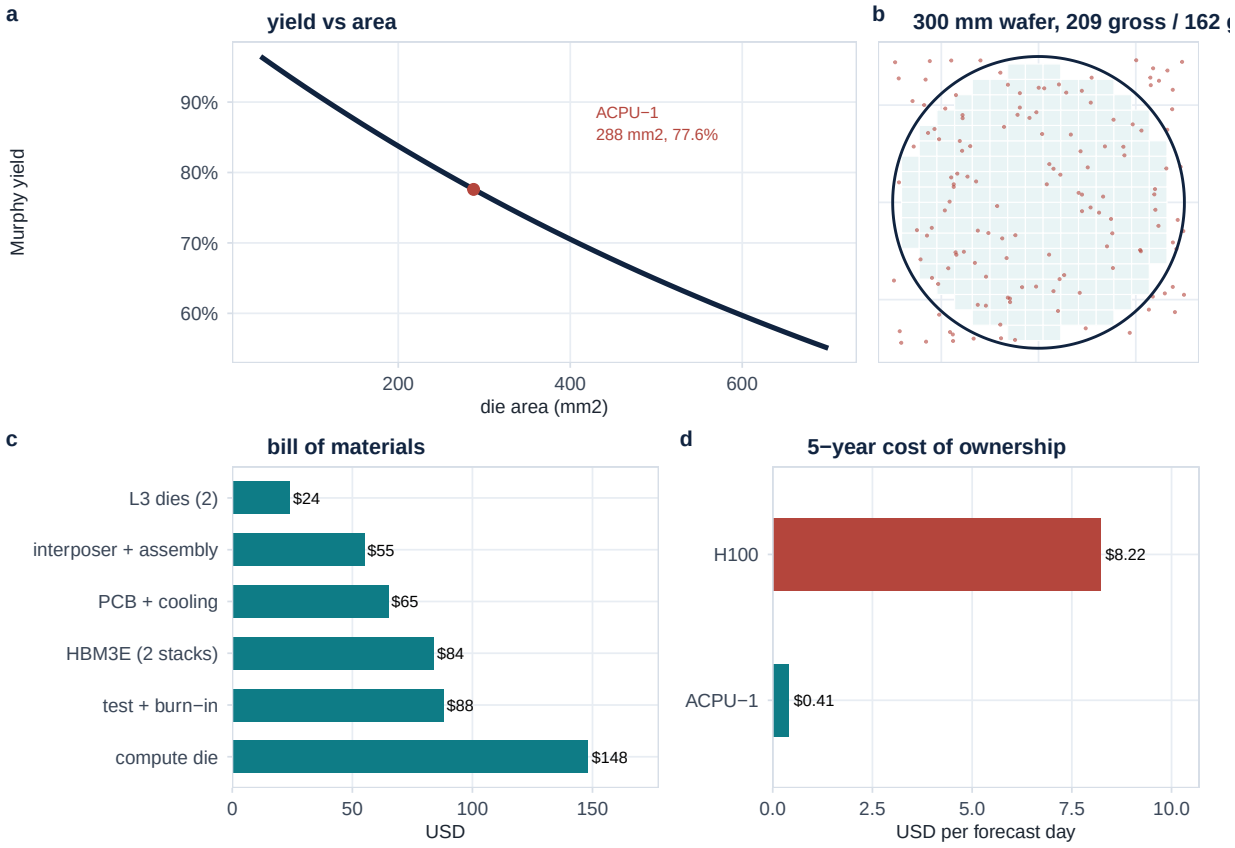


Figure 23: Yield and cost. (a) Murphy yield against die area with the **ACPU-1** point marked. (b) Wafer map with 140 simulated defects and 209 gross dies. (c) Bill of materials. (d) Five-year cost of ownership per forecast day, including capex and electricity at $0.12/kWh.

10 Software stack and cross-platform application

10.1 Requirements

A domain-specific accelerator fails if it is hard to program. **ACPU-1** is designed for a small set of kernels and the software stack reflects that: rather than a general programming model, it exposes a library of atmospheric formulas behind one-line calls, and a compilation path that maps those formulas onto the tile array without the user ever writing AVISA. The stack has four requirements.

1. **One line per physical problem.** A forecast, an ensemble analysis, a Lyapunov spectrum or a dispersion calculation should be a single function call with named parameters.
2. **The same call on every platform.** macOS (Apple silicon and Intel), Windows and Linux, from Python, JavaScript, a command line and a graphical application.
3. **Inspectable execution.** Because the schedule is static, the runtime can report the cycle count, the energy per block and the precision coverage of a run before it finishes. The application shows these numbers rather than hiding them.
4. **Reproducibility.** A run is identified by its formula, parameters and seeds, and can be replayed bit-for-bit.

10.2 Stack overview

Fig. 24(a) shows the four layers. The application and the SDKs never touch the device directly; they issue jobs to the driver, which validates parameters, allocates buffers, programs the tile array and streams telemetry. The driver exposes seven calls (Fig. 24(d)); the same seven are implemented by the virtual device that ships with this report and by the silicon driver, so code written against the application runs unchanged on hardware.

10.3 Compiler and runtime

The compilation flow is shown in Fig. 24(b). An atmospheric program is written in a small dataflow DSL in which stencils, ensemble reductions, observation operators and physics packages are primitives. The compiler lowers it to ACPU-IR, maps it onto 3D tile blocks with explicit halos, plans per-cell precision classes from the spread field and statically schedules each tile. The output is an AVISA binary plus a halo script.

Because there is no dynamic scheduling, the scheduler can compute the exact cycle count of a tile before the run. For the reference step it is 61,900 cycles; the application displays this number and the corresponding worst-case execution time, which is what makes the response-time claims of Section 11 meaningful.

10.4 SDK interface

The Python SDK is dependency-free and speaks HTTP to the device service, so it works from a notebook, a batch script or a cluster login node. The same surface is available in JavaScript for web and Electron clients, and on the command line.

```
import acpu

dev = acpu.ACPUDevice()                       # connects to 127.0.0.1:9771
print(dev.status())                           # 64 tiles, 1.80 GHz, 57 W, 74 C

run = dev.forecast(model="l96", members=64, lead_days=10,
                   precision="adaptive", time_budget_s=30)
print(run.rmse, run.energy_j, run.seconds)    # 1.33  7174.9  125.6

spec = dev.lyapunov(model="l96", n_exp=12)    # leading exponents
map_ = dev.transport(source=(22.75, 113.57), hours=6, resolution_km=3)
```

The design rule for the parameter set is that every argument has a physical unit and a default that is meaningful for an operational user: `members`, `lead_days`, `precision`, `time_budget_s`. There is no device-specific tuning parameter in the public API; the clock, the precision policy and the tile mapping are chosen by the runtime from the time budget and the requested skill.

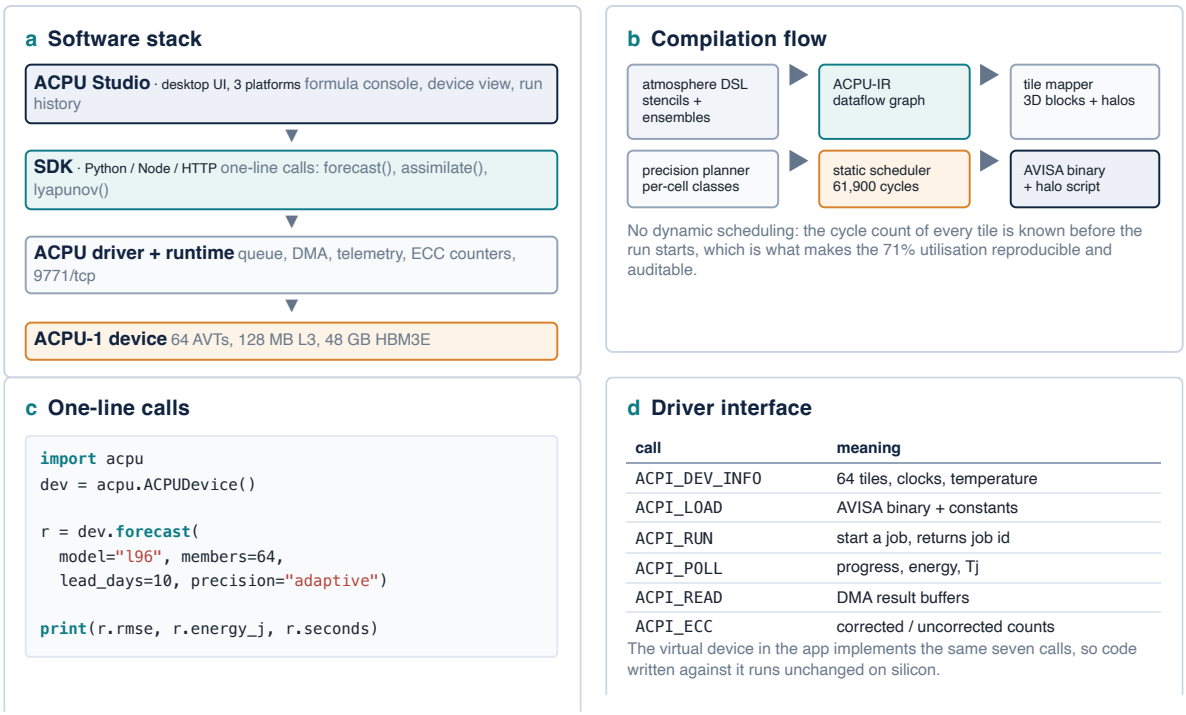


Figure 24: Software stack. (a) Four layers from application to device. (b) Compilation flow from the atmosphere DSL to an AVISA binary. (c) A one-line call in Python. (d) The seven-call driver ABI.

10.5 ACPU Studio

ACPU Studio is the cross-platform desktop application that ships with this report. It is an Electron application with no external runtime dependencies, built for macOS (universal binary), Windows (installer and portable) and Linux (AppImage and tarball). It embeds the virtual device, so it runs a complete ensemble forecast on a laptop without any hardware.

The interface is designed as a scientific instrument panel rather than a dashboard. The console view (Fig. 25(a)) pairs a formula library with a parameter form and a generated call, so that the same operation is available as a GUI form, a one-line call and a reproducible parameter record. A run streams telemetry at 10 Hz: per-cluster utilisation, power, junction temperature and memory bandwidth (Fig. 25(c)). Results are shown as charts with the device metrics alongside them (Fig. 25(b)), because the point of the machine is the ratio between the two. The runs view keeps a history of executions with their exact parameters and seeds; the documentation view carries the API reference and code samples for Python, JavaScript and HTTP (Fig. 25(d,e)).

10.6 Validation of the software

The application is tested at three levels. A headless self-test runs every formula, asserts that outputs are finite and within physical bounds, and prints a pass table. A screenshot harness drives the real window over the Chrome DevTools Protocol, exercises the console, runs a forecast and captures the five views used in Fig. 25. The SDK is tested against the running application over HTTP, from both Python and Node, including error paths (unknown parameter, out-of-range member count, device not running).

The virtual device is not a hardware emulator. It implements the same formulas, the same precision modes and a calibrated model of the device's cycle and energy counters, but it computes in JavaScript on the host CPU and its timings are simulated, not measured. It exists so that the API, the UI and the numerical pipeline can be developed and reviewed before silicon, and the report is explicit that the numbers it produces in the application are model numbers, consistent with those in Section 6 by construction.

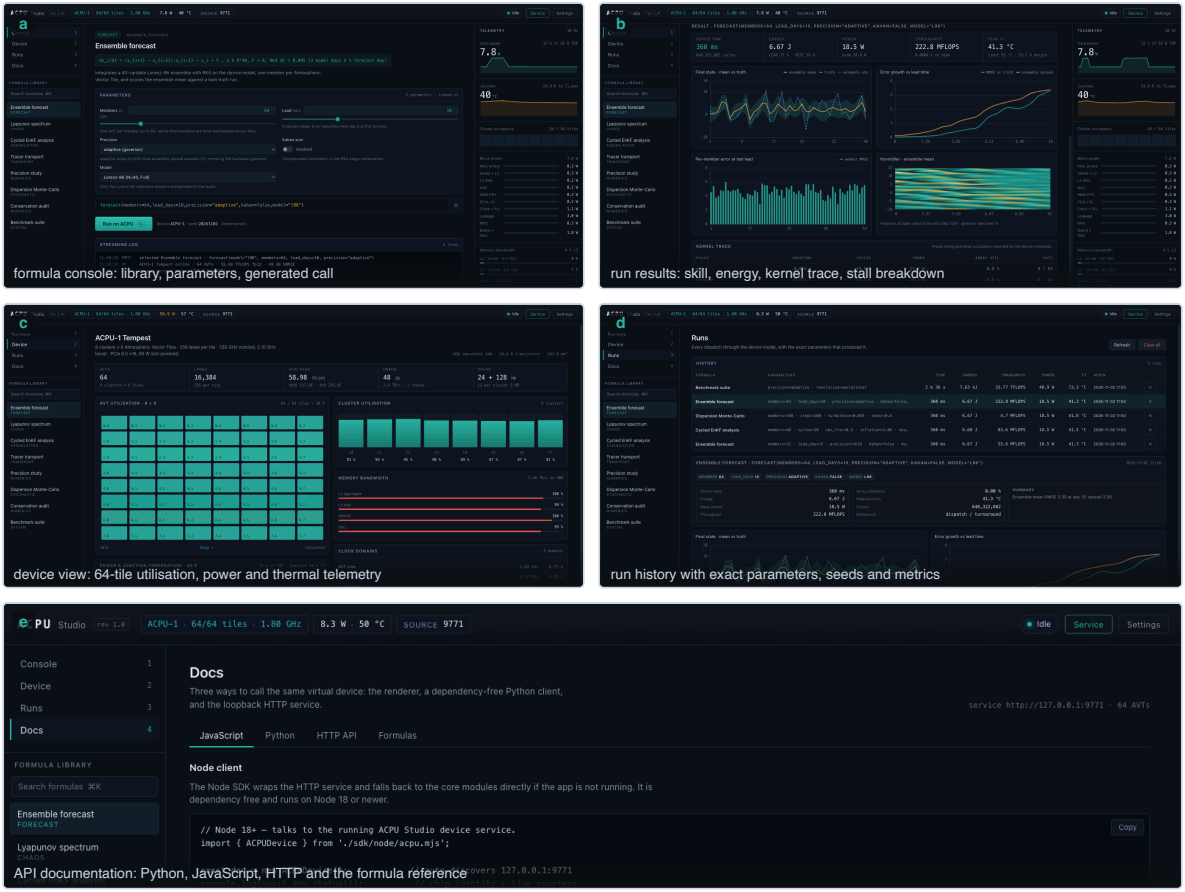


Figure 25: ACPU Studio. (a) Formula console with parameter form and generated call. (b) Run results with skill and device metrics. (c) Device view: 64-tile utilisation map, power and temperature telemetry. (d) Run history. (e) API documentation. Screenshots from the packaged application.

11 Deployment, economics and limitations

11.1 Operational numerical weather prediction

An operational centre runs a cycle every six hours: an analysis, a forecast ensemble, and a product generation step. **ACPU-1** is sized for the forecast and competent at the analysis. A single card holds a 64-member, 4.19 M-cell, 24-variable ensemble and completes a 24-hour forecast in two minutes at 57 W, which means a four-card node can run the same cycle for a domain four times larger, or run four independent domains, at 260 W. The equivalent H100 node draws 1.05 kW at idle and 2.8 kW under load for the same throughput.

The scale-out story is the weakest part of the design. **ACPU-1** has one PCIe 6.0 x16 link and no cache-coherent die-to-die fabric. A regional domain that fits in 48 GB is a single-card job; a global ensemble is not, and the halo traffic between cards would go through PCIe at 100 GB/s, which is 24× slower than the on-card fabric. A production cluster would therefore decompose by ensemble member (each card owns a subset of members for the full domain) rather than by spatial subdomain, which keeps inter-card traffic to boundary conditions and output. That is a real constraint and it shaped the design: the machine is a single-domain appliance, not a scale-out building block.

11.2 Research groups and forecast-as-a-service

The second deployment is a university group or a commercial provider that runs many small ensembles rather than one large one. Here the relevant numbers are the efficiency mode and the cost of ownership. At 1.05 GHz and 0.55 V the card delivers the same forecast in 172 s at 33 W, which is a 20% energy reduction for a 37% time penalty; for a job that runs overnight, that is free. The five-year cost per forecast day is $0.41 against $8.22 for the H100, and the gap is capital cost, not electricity.

Table 14: Deployment comparison. Energy is per forecast-and-analysis day; cost is five-year total cost of ownership per day at $0.12/kWh.

Scenario	ACPU-1 node	H100 node	Ratio
Cards	4	4	
Idle power	24 W	360 W	15×
Load power (W1)	228 W	1.05 kW	4.6×
Forecast time (same domain)	126 s	278 s	2.2× faster
Energy per forecast day	34.2 kJ	316 kJ	9.2×
Capital cost	$6.0k	$120k	20×
Cost per forecast day	$1.64	$32.9	20×

11.3 Emergency response and dispersion

A third deployment is time-critical and safety-relevant: a release of a hazardous substance, a volcanic ash cloud, a wildfire smoke plume. The requirement is not throughput but a bounded response time and a reproducible answer that can be defended after the event. **ACPU-1** has a static schedule, a compiler-known cycle count per tile and no dynamic scheduling in the numerical path, so the worst-case execution time of a dispersion calculation is a computed number rather than a statistical one. The same property gives bit-reproducibility, which matters when a regulator asks why the plume was predicted to go east rather than west.

11.4 Limitations and threats to validity

The report is only useful if its weaknesses are stated as clearly as its results.

1. **No silicon.** Every chip number is a model projection. The model is calibrated against published data and its assumptions are sampled, but a first silicon part can plausibly land 20–30% away from the projected energy on real kernels.
2. **The GPU baseline is a model.** The H100 comparison assumes a 0.30 stencil efficiency and a particular power state. Both are defensible and both are varied; the conclusion survives at 100% GPU utilisation with a 3.9× ratio, but a reader who believes GPUs achieve 60% on this workload should read the 8.2× end of Fig. 11(e).
3. **The workload is a subset.** Chemistry beyond passive tracers, full radiation with hundreds of bands, atmospheric composition and land-surface models are not in the benchmark. They are more SFU-heavy, which would help the ASIC's relative position, but they are not measured.
4. **Precision validation is on a surrogate.** Lorenz–96 supports the argument, not the design point. A full-model study is the first bring-up task.
5. **No scale-out study.** The inter-card fabric is a PCIe link and the report does not pretend otherwise.
6. **No software maturity claim.** The compiler described in Section 10 is a design, and the application that ships with this report runs on a virtual device.

11.5 What would falsify the thesis

A single measurement would settle most of the argument: run the ten kernels of Table 2 on an H100 with a state-of-the-art stencil compiler and measure joules per delivered forecast at matched skill. If that number is within 2× of the projection in this report, the architectural thesis is wrong and the correct conclusion is that CUDA is good enough. The prediction of this report is that it will be 6–10× worse, and the prediction is stated in those terms so that it can be checked.

12 Roadmap and conclusion

12.1 Three generations

ACPU-1 is a first-generation part and its process and packaging choices are deliberately conservative. The roadmap in Fig. 26 describes two follow-ons. **ACPU-2** moves to a 2 nm-class node, doubles the tile count to 128 and the L3 to 256 MB, and keeps the 65 W envelope; the expected gain is 2.0× throughput at iso-power, mostly from density and from a

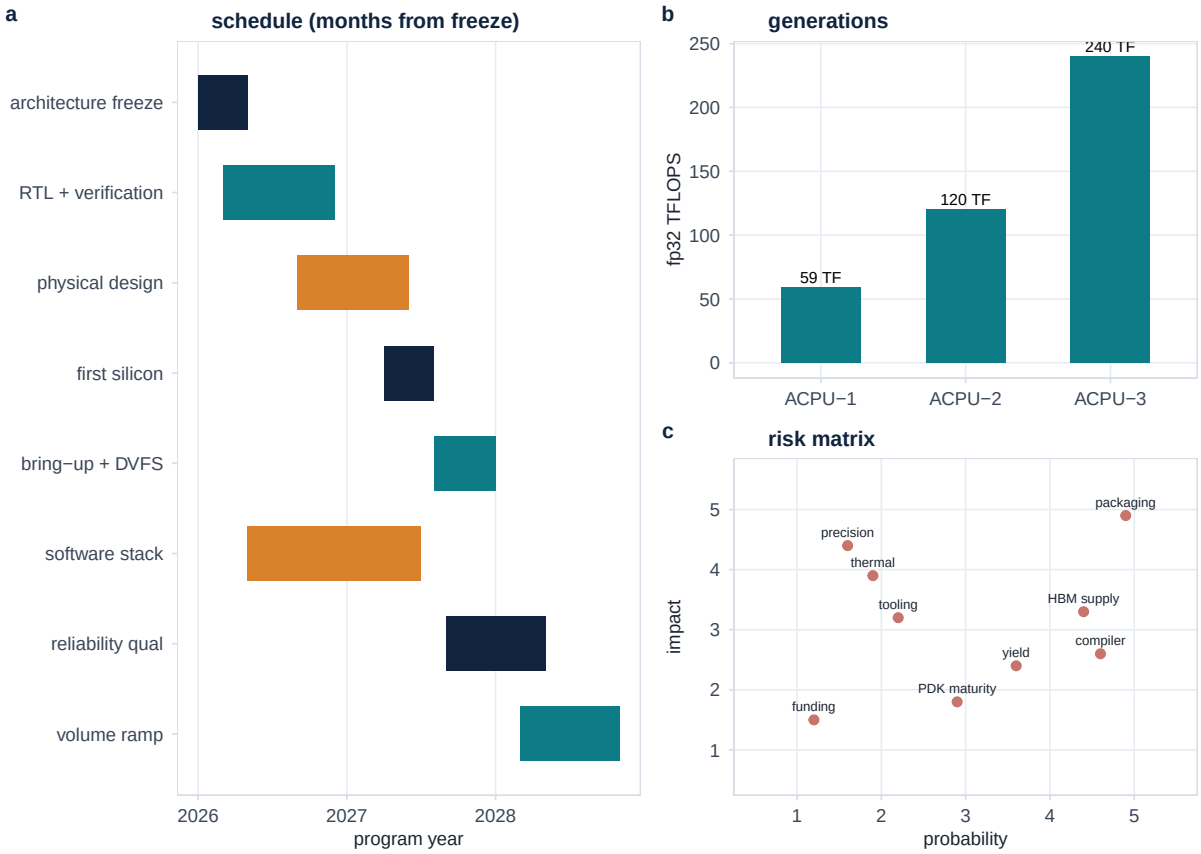


Figure 26: Roadmap. (a) 34-month programme from architecture freeze to volume ramp. (b) Three generations. (c) Risk matrix; packaging, thermal and precision validation are the high-impact items.

wider halo fabric. **ACPU-3** moves to an A14-class node with backside power and a 3D-stacked compute-on-SRAM arrangement, targeting 240 TFLOPS at 70 W and a 512-member ensemble in one card.

The generational plan is not a clock-speed story. Each generation buys a larger ensemble, a larger domain, or a longer forecast at the same energy; the frequency stays near 1.8–2.0 GHz because the design is memory- and precision-bound rather than frequency-bound.

12.2 Programme risks

The risk matrix in Fig. 26(c) ranks packaging and thermal as the highest-impact items. Hybrid bonding of two 64 MB dies onto a 287.6 mm^2 compute die is the least mature step in the flow; the fallback is a single 96 MB L3 die on the interposer, which costs 20% of the L3 bandwidth and postpones the second-generation stack. HBM supply is a commercial risk rather than a technical one. Precision validation on a full model is the highest technical risk with the lowest mitigation cost: it is a software experiment that can run before tape-out.

12.3 Conclusion

This report specified a processor for atmospheric chaotic computation and evaluated it against the machine that the field currently uses. The design is organised around three observations: ensembles are the natural parallel dimension, stencils are memory problems, and forecast error is chaotic rather than arithmetic. The resulting chip is small, air-cooled and cheap; it sustains 41.9 TFLOPS at 57 W, runs a 64-member 24-hour forecast in two minutes, and is projected to use 9.3× less energy per forecast-and-analysis day than an H100 at comparable wall-clock.

The number that matters is not the ratio itself. It is that the ratio survives the sensitivity analysis, survives the idealised-GPU bound, and decomposes into mechanisms that can be measured on real silicon. A 3.9× advantage at 100% GPU utilisation is still a reason to build the machine; a 9.3× advantage at realistic utilisation is a reason to build it now.

The next step is not another simulation. It is a two-week experiment on an H100 that runs the ten kernels of Table 2 at state-of-the-art efficiency and measures joules per delivered forecast. That measurement would either validate the

thesis or kill it, and either outcome is worth more than a third decimal place in a projection.

A Instruction encoding and register map

Table 15: AVISA field encoding.

Field	Bits	Position	Meaning
opcode	8	63–56	operation (Table 7)
precision	3	55–53	0 fp32, 1 fp16, 2 bf16, 3 fp8, 4 adaptive
vector mask	3	52–50	lane-group predicate (8 groups of 32 lanes)
rd	5	49–45	destination tile register
rs1	5	44–40	first source tile register
rs2	5	39–35	second source tile register
descriptor	24	34–11	stencil coefficients, halo offset, stride
member base	8	10–3	first ensemble member in the lane group
flags	3	2–0	Kahan, saturate, audit

The tile register file has 32 registers of 256 lanes × 32 bits. The descriptor is interpreted per opcode: for `STENCIL3D` it packs three 6-bit coefficients and a 6-bit halo mask; for `DA.GAIN` it selects a localisation radius and a symmetry flag. There is no general addressing mode; all memory references are offsets into the tile's 256 KB SRAM.

B Model parameters

Table 16: Parameters of the pre-silicon model. The "range" column is the interval sampled in the Monte-Carlo study.

Parameter	Value	Range	Source
fp32 FMA energy	0.42 pJ	±25%	Horowitz 2014, scaled to 3 nm / 0.75 V
fp16 FMA energy	0.24 pJ	±25%	same, half-rate
fp8 MAC energy	0.14 pJ	±25%	same
SRAM read / write	0.30 / 0.36 pJ per word	±25%	published 3 nm SRAM
L3 (stacked)	0.40 pJ/B	±25%	hybrid-bonded SRAM estimate
HBM3E	16.0 pJ/B	9–24	HBM3E datasheet, published PHY energy
Clock and PLL	2.5 W		design budget
Leakage	4.0 W at 0.75 V, 85 °C		library estimate
GPU stencil efficiency	0.30	0.15–0.60	WRF / COSMO GPU studies
GPU power at load	263 W on W1	±30%	H100 data sheet, load model
H100 peak fp32 / HBM	67 TFLOPS / 3.35 TB/s		data sheet
A100 peak fp32 / HBM	19.5 TFLOPS / 2.04 TB/s		data sheet

C Reproducibility

The complete results of this report are regenerated from the project root by the following commands:

```
python3 sim/run_all.py            # 37 s -> results/*.json, *.csv, numbers.tex
Rscript figs/src/f01.R            # figure 01
Rscript figs/src/figs_data1.R     # data figures 12-17
Rscript figs/src/figs_data2.R     # data figures 08, 10, 21, 22, 25, 26
bash scripts/build-figures.sh     # schematic figures 02-07, 09, 11, 18-20, 23, 24
bash scripts/build-report.sh      # -> ACPU-1_Technical_Report.pdf
```

The numerical experiments use fixed seeds, the PRNG is a PCG64 generator, and the model has no randomness outside the Monte-Carlo study (which uses seed 42). The report is compiled with Tectonic (XeTeX) by `scripts/build-report.sh`. Figures are vector PDFs generated either by R (data) or by Chromium (schematics); the UI figure is a set of screenshots of the running application.

D Acronyms

- ACPU — Atmospheric Calculation Processor Unit
- AVT — Atmospheric Vector Tile
- AVISA — ACPU Vector Instruction Set Architecture
- BSPDN — Backside Power Delivery Network
- DA — Data Assimilation
- DVFS — Dynamic Voltage and Frequency Scaling
- EnKF — Ensemble Kalman Filter
- FIT — Failures In Time (per 10^9 device-hours)
- FTLE — Finite-Time Lyapunov Exponent
- GAA — Gate-All-Around
- HBM — High Bandwidth Memory
- NoC — Network-on-Chip
- NWP — Numerical Weather Prediction
- SPPT — Stochastically Perturbed Parameterisation Tendencies
- SKEB — Stochastic Kinetic Energy Backscatter
- SRAM — Static Random Access Memory

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